

From Recordings to Ecology

Turning Bird Sounds into Ecological Insights

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2026 April 1st, PhD defence





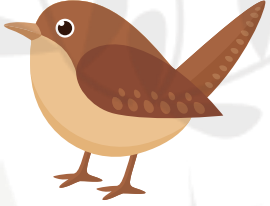
Acoustic Data

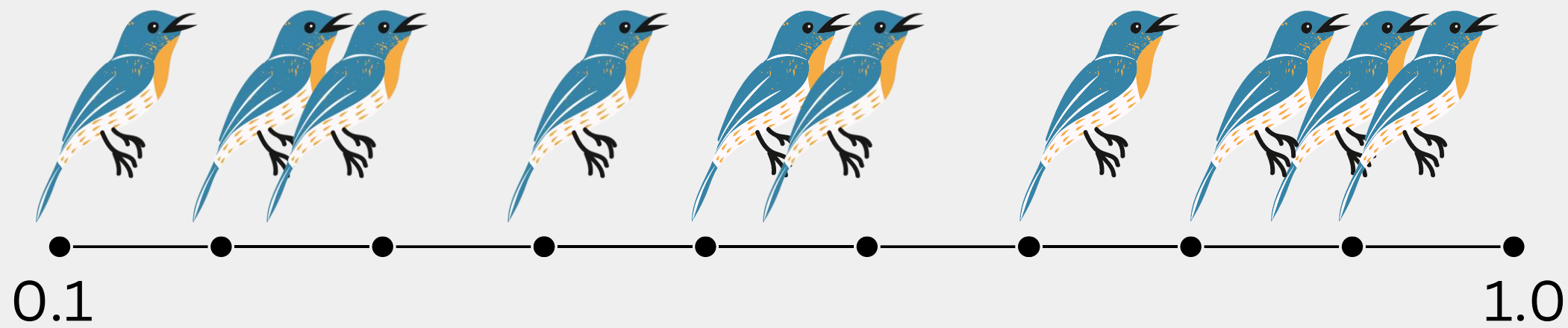


BirdNET



Species Detections

Species	Confidence value
	$0 \sim 1$





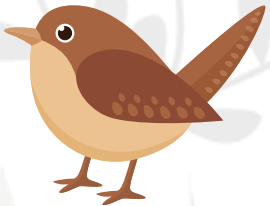
Acoustic Data



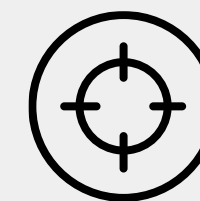
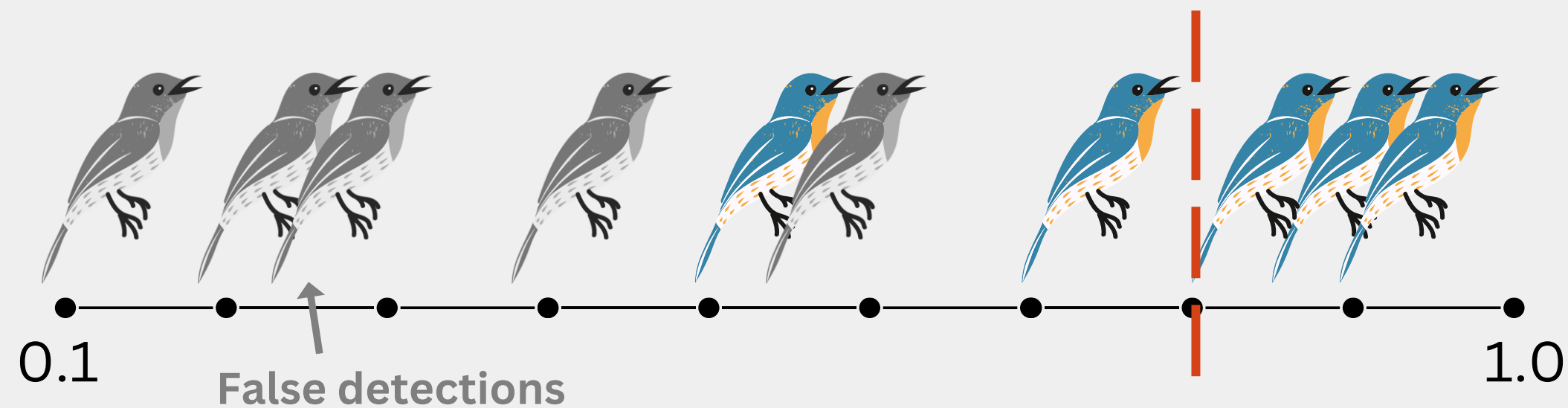
BirdNET



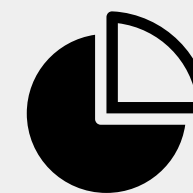
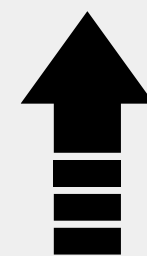
Species Detections

Species	Confidence value
	$0 \sim 1$

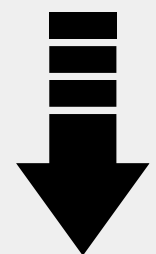
High threshold



Precision



Data retained





Acoustic Data



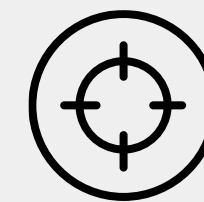
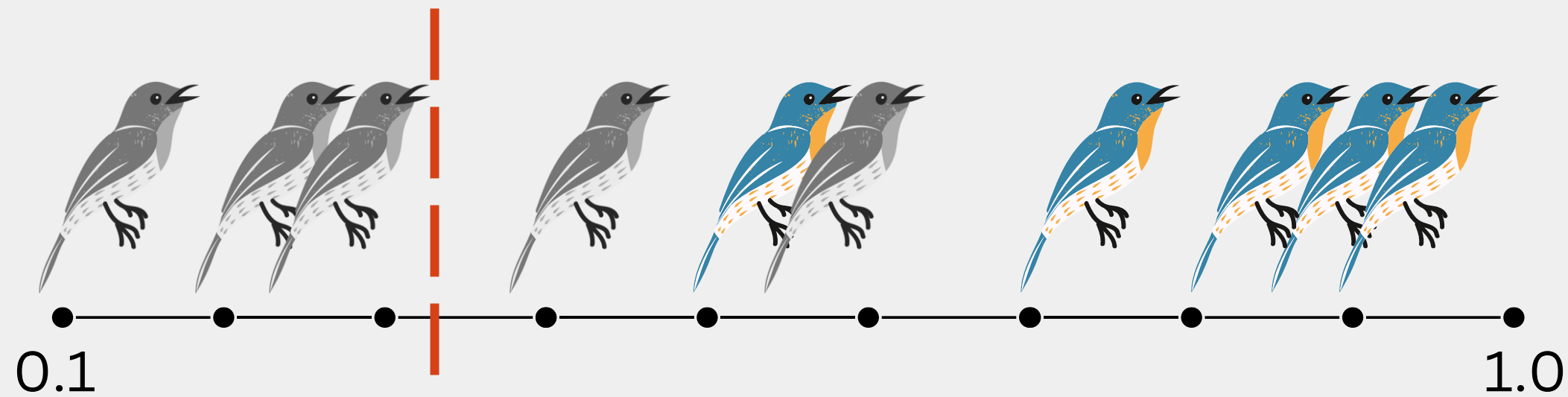
BirdNET



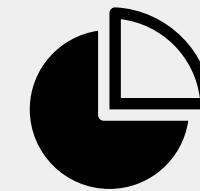
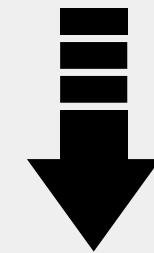
Species Detections

Species	Confidence value
	$0 \sim 1$

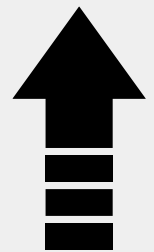
Low threshold



Precision

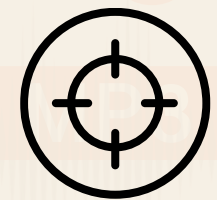


Data retained

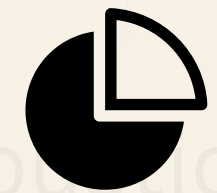


Species

Confidence value



Precision



Data retained



Species-specific

**A framework to define species-specific thresholds for
BirdNET's optimal performance**

0.1

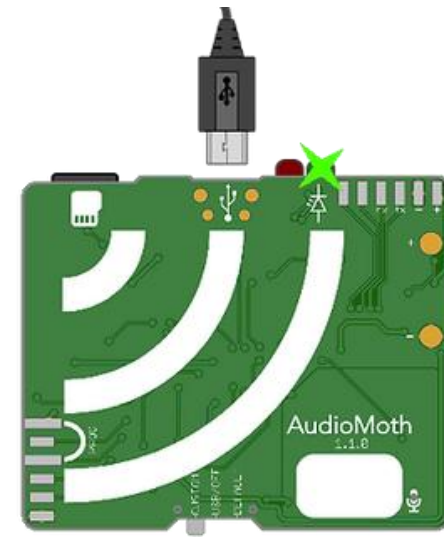
1.0

Data retained



- 66 ARUs
- ~3 km apart

Location



AudioMoth

- 3 summers (May - Jul, 4 - 7am)
- 2 winters (Feb - Apr, 6 - 10pm)
- > 10,000 hrs

Time

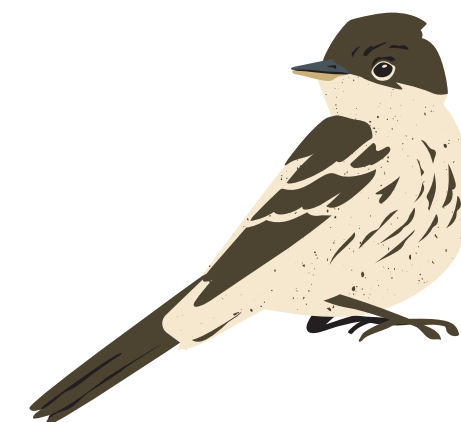
19 target species



6 warblers



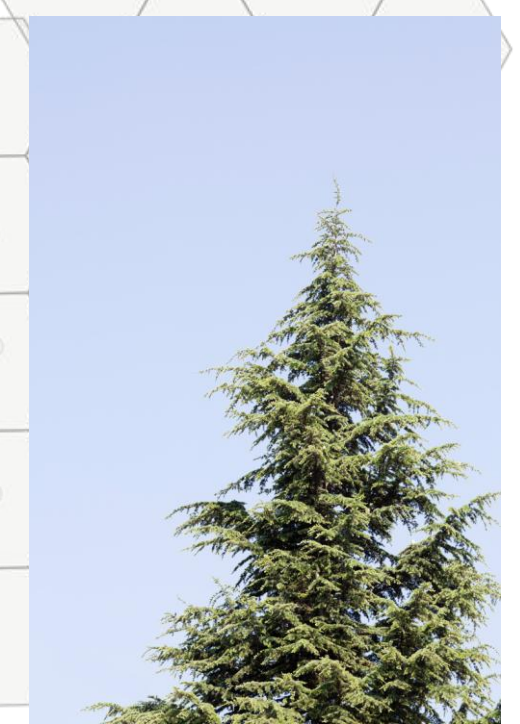
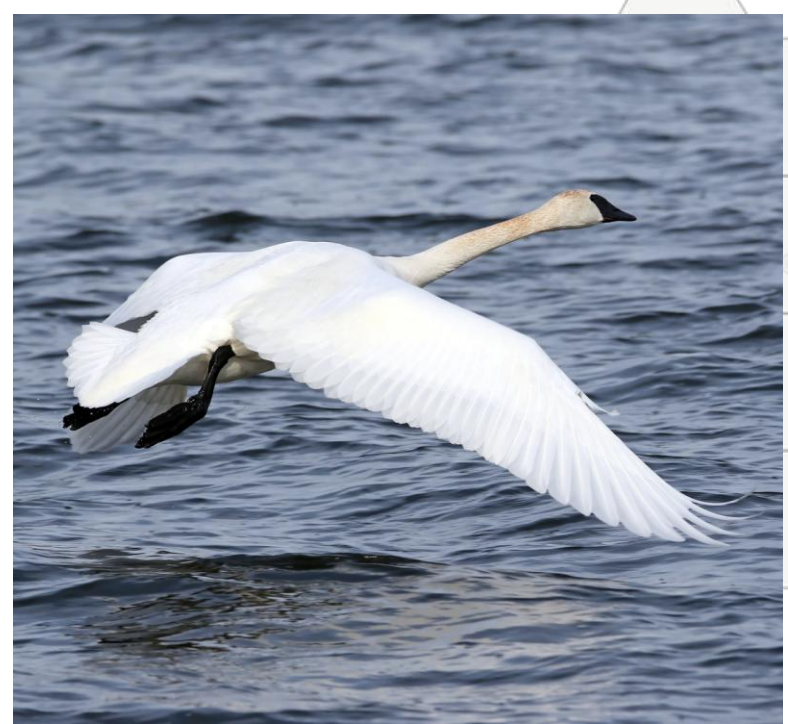
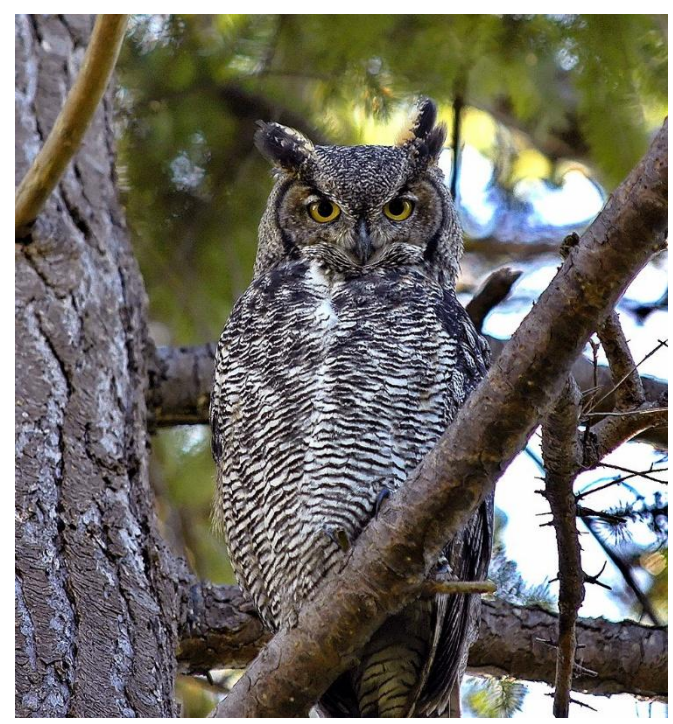
3 thrushes



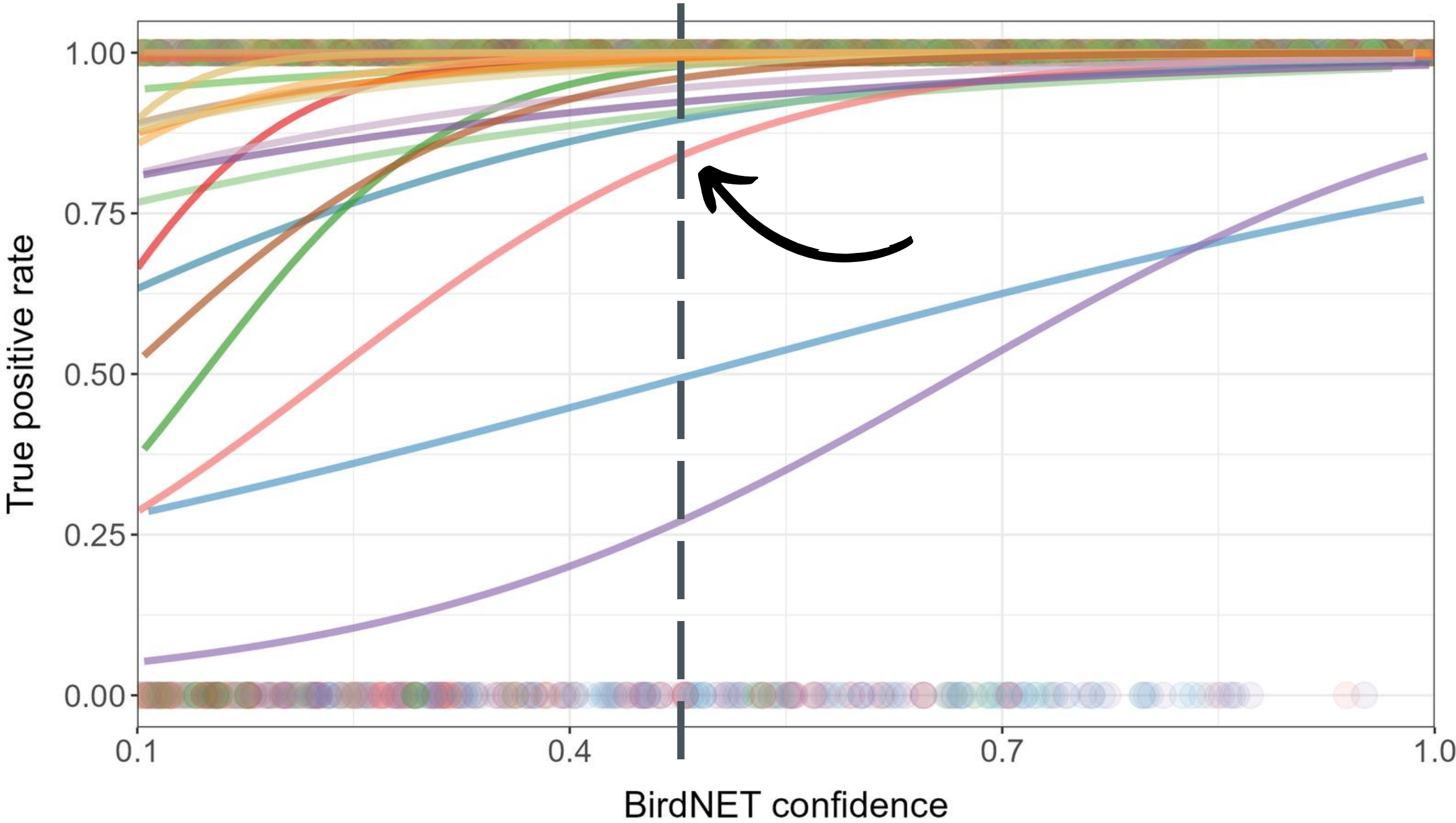
4 flycatchers



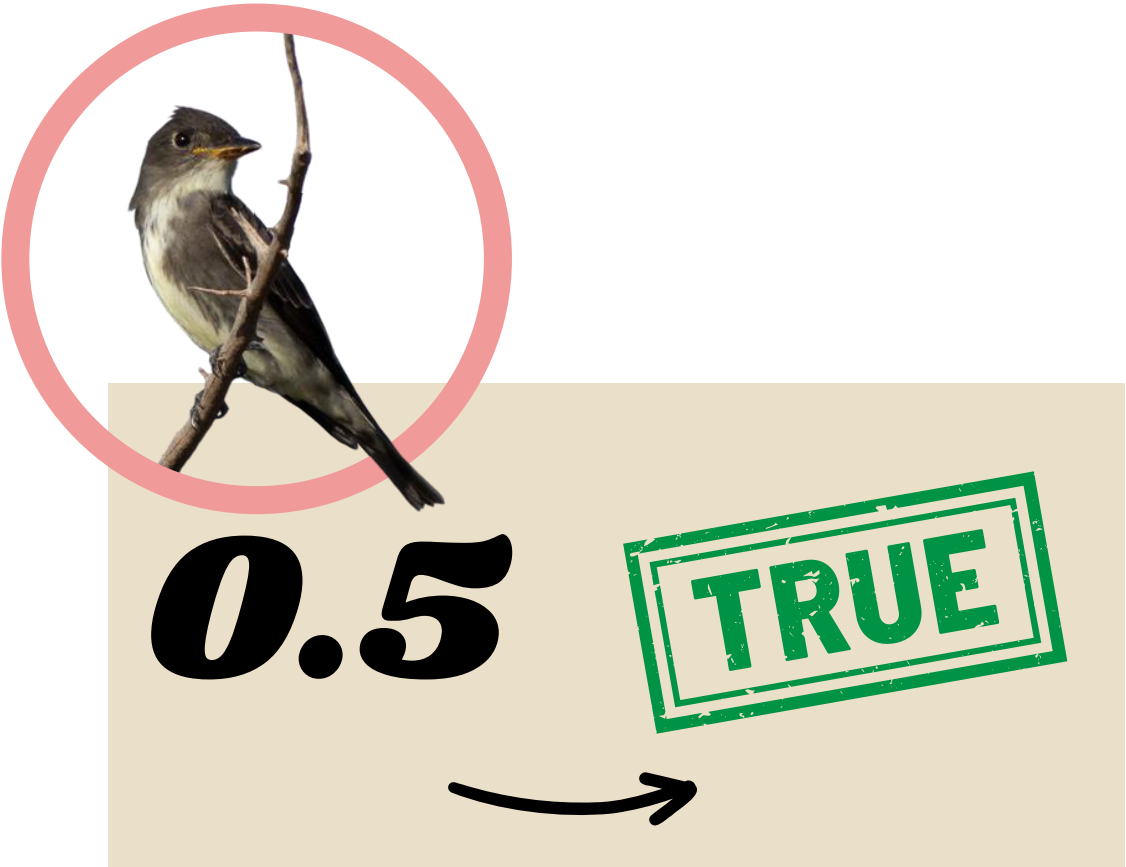
6 other



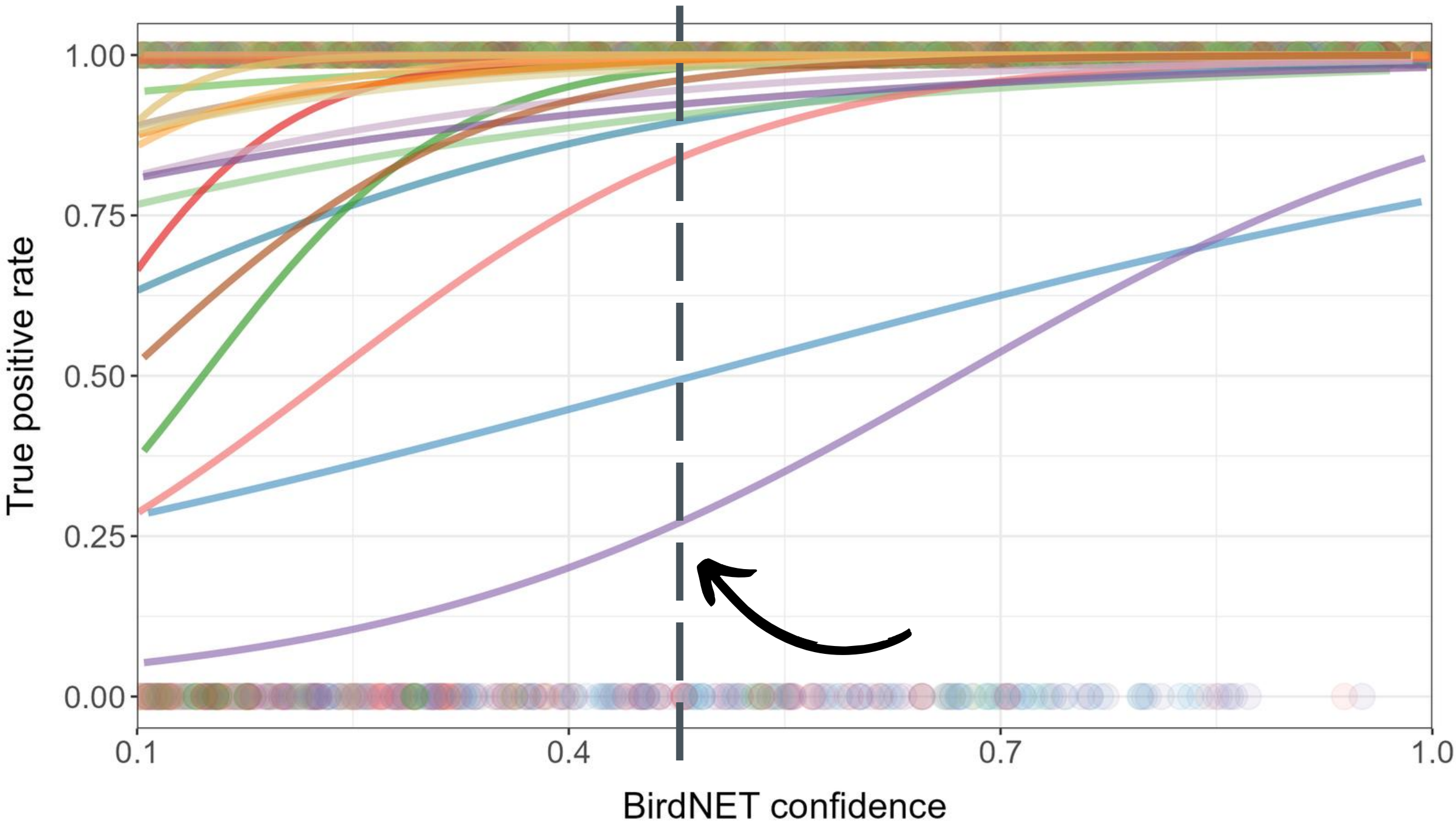
Translating BirdNET confidence to probability



The meaning of the confidence value differs by species

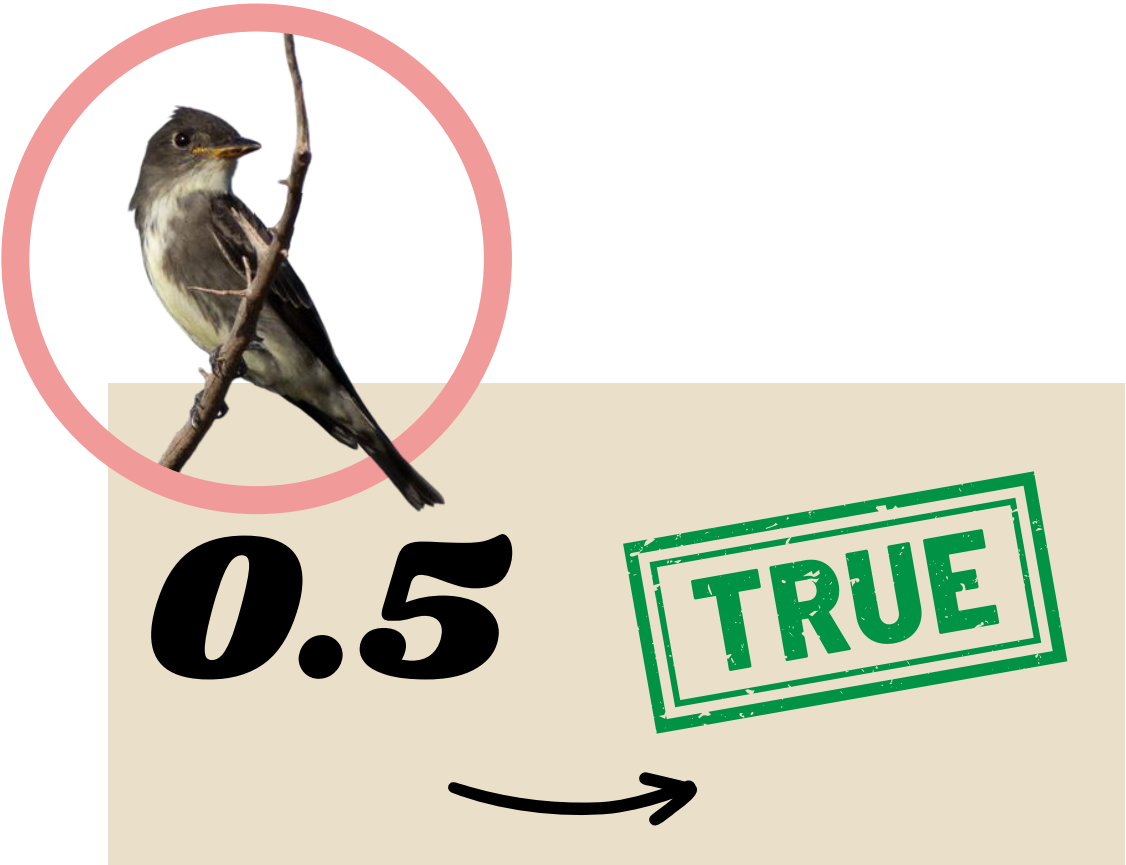


Translating BirdNET confidence to probability

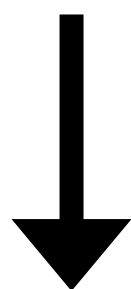


- | | | | |
|------------------------|------------------------|-----------------------|------------------------|
| American Robin | Northern Flicker | Red-breasted Nuthatch | Western Flycatcher |
| Brown Creeper | Northern Waterthrush | Ruby-crowned Kinglet | Western Wood-Pewee |
| Hammond's Flycatcher | Olive-sided Flycatcher | Swainson's Thrush | White-throated Sparrow |
| MacGillivray's Warbler | Orange-crowned Warbler | Tennessee Warbler | Wilson's Warbler |
| Myrtle Warbler | Pacific Wren | Varied Thrush | |

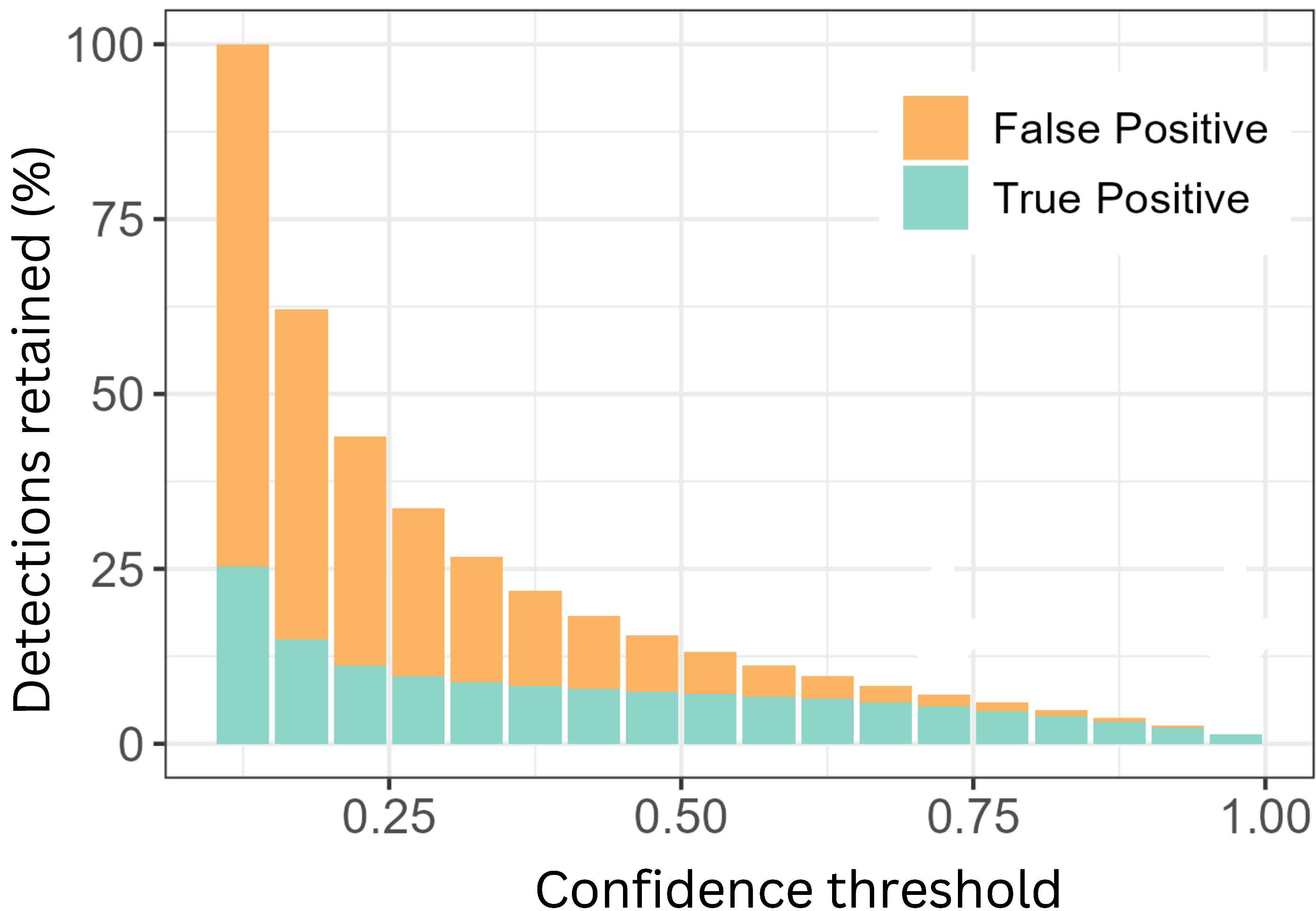
The meaning of the confidence value differs by species



Threshold



**Data retained
Precision**

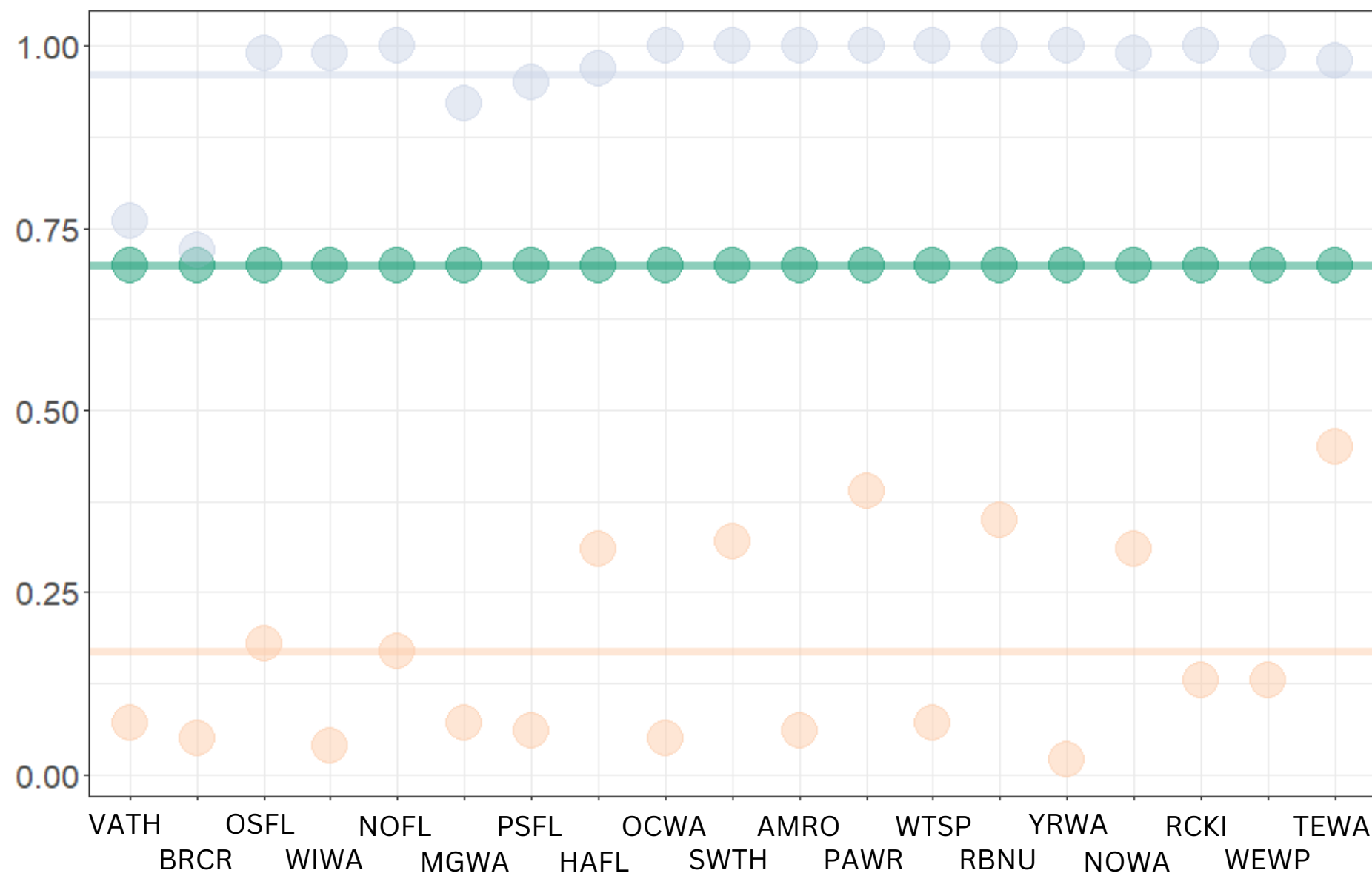


Confidence threshold setting

- **Precision:** High precision in both methods, ranging from 0.7 to 1
- **Data retained:** 17% for universal threshold; 71% for species-specific
- **Threshold:** Species-specific threshold is generally lower than 0.25

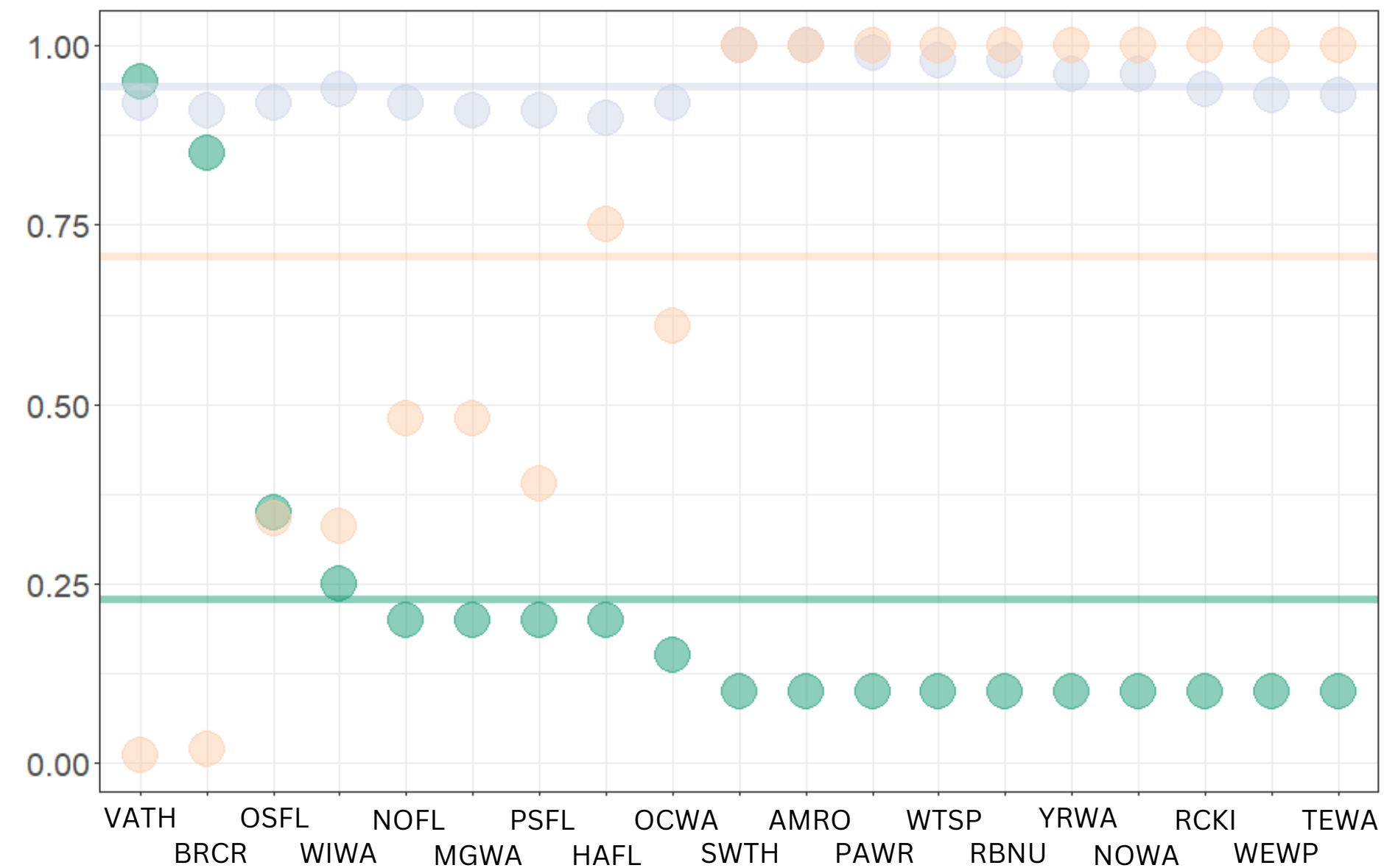
Universal threshold

A general high threshold of 0.7



Species-specific threshold

A minimum threshold to achieve 0.9 precision

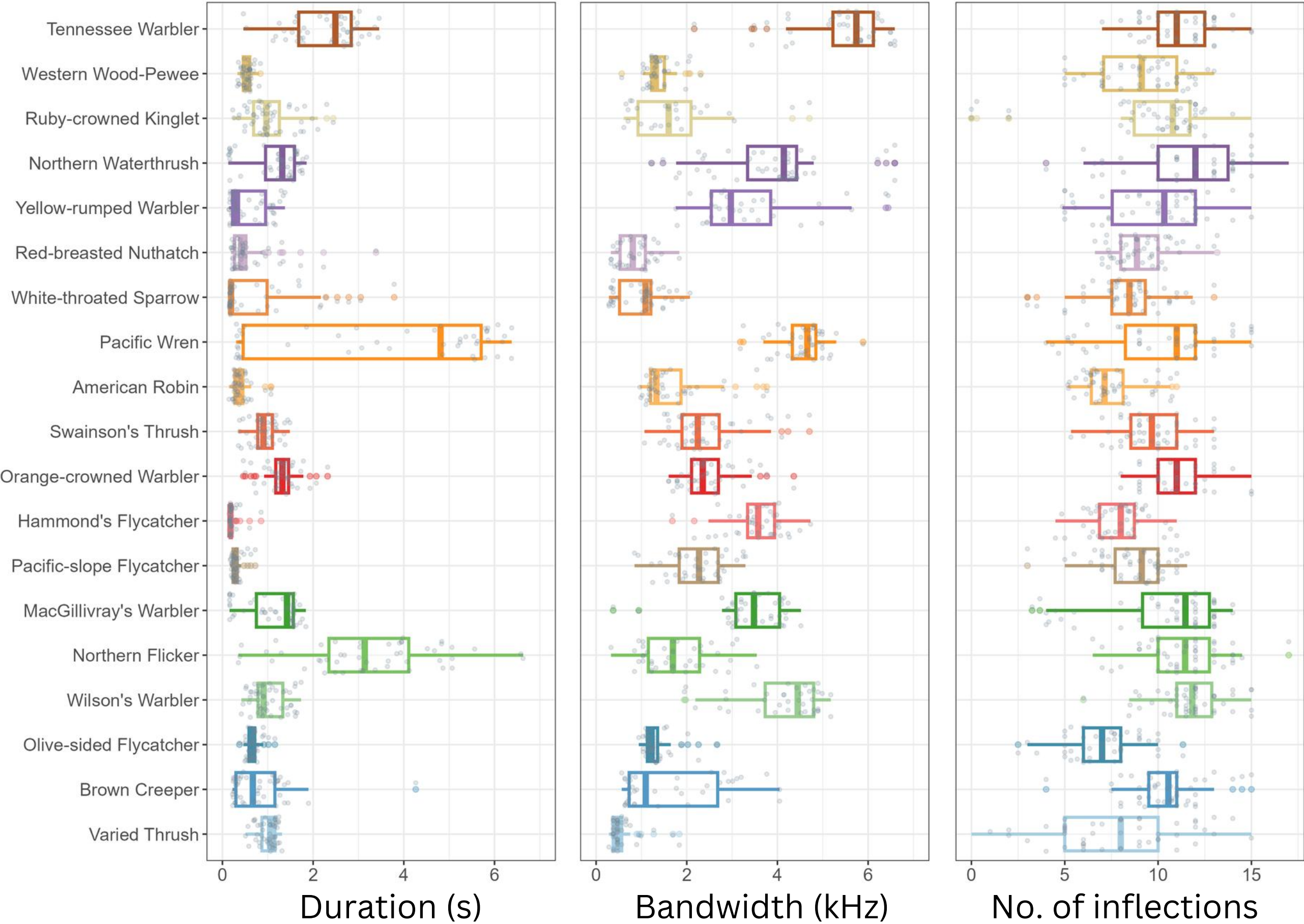


Is species-specific threshold related to vocal complexity?

Low threshold

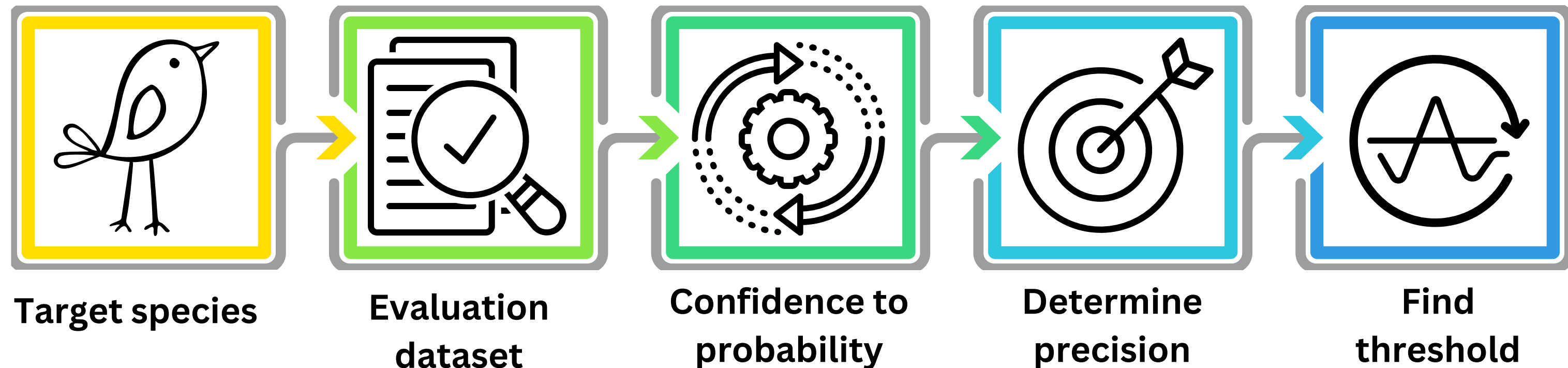


High threshold



Setting BirdNET confidence thresholds

1. **BirdNET** is a powerful tool
2. **Species-specific threshold** provides higher precision & data retained
3. Species thresholds are generally low, **no clear link with vocal complexity**



We provided a framework of defining BirdNET species-specific thresholds to ensure reliable output for passive acoustic monitoring.

Machine learning output → ecological insight

Within species



**Identifying
individual owls**

Single species



**Finding spatiotemporal
vocal activity patterns**

Multiple species



**Linking habitat
features to richness**

Machine learning output → ecological insight

Within species



Identifying
individual owls

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Finding spatiotemporal
vocal activity patterns

Multiple species



Linking habitat
features to richness

Individual identification

Why

- Social hierarchy
- Resources use and physical development
- Individual territory

How

- Physical tags
- GPS tracking
- Physical features



Active recording



Passive recording?

Evaluating the potential of individual identification of Barred Owls using passive acoustic monitoring

Captures of animal & visual cue

© Chiara Fumagalli

© Jasmin Pape

Barred Owl vocalizations

Territory establishment

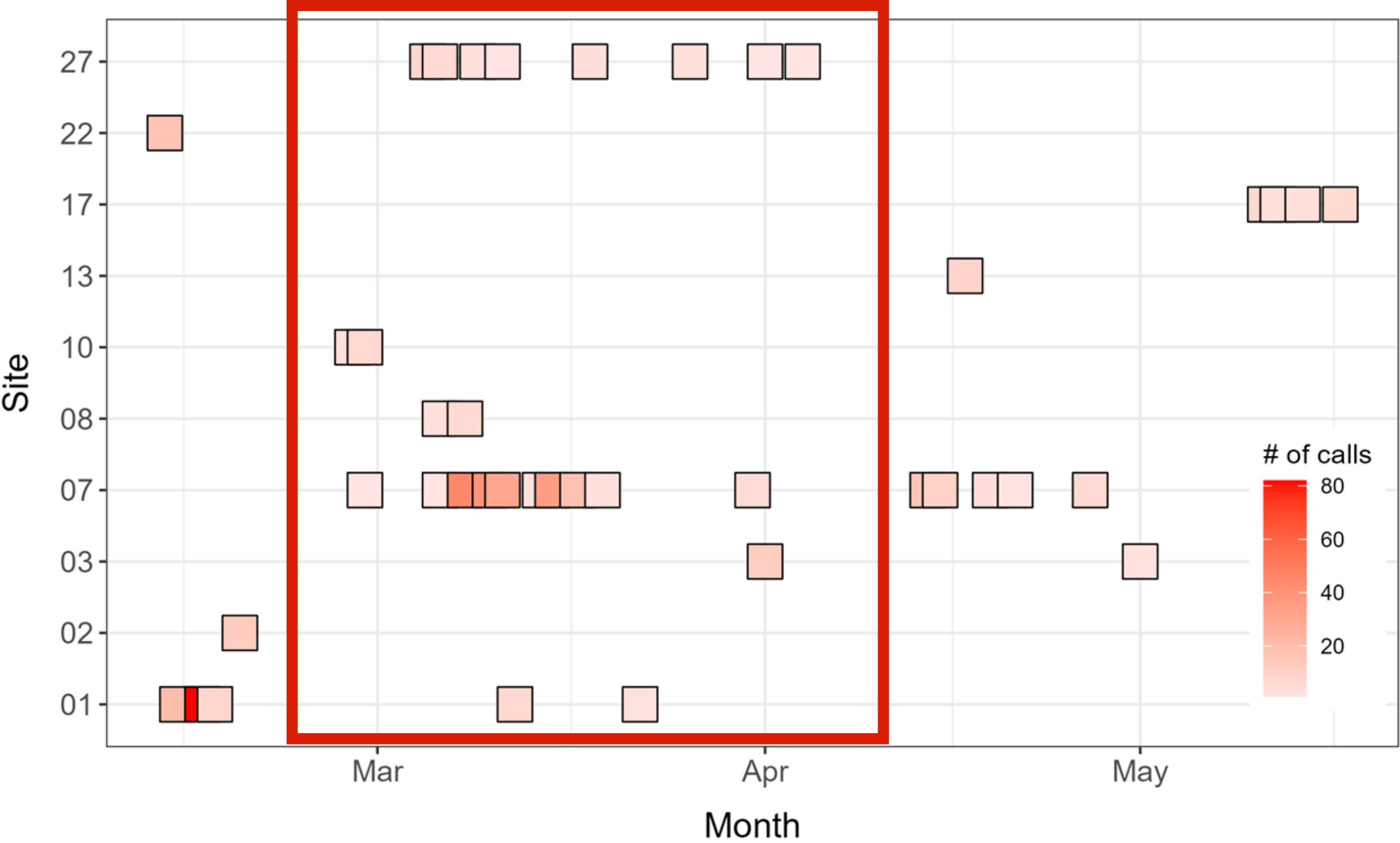
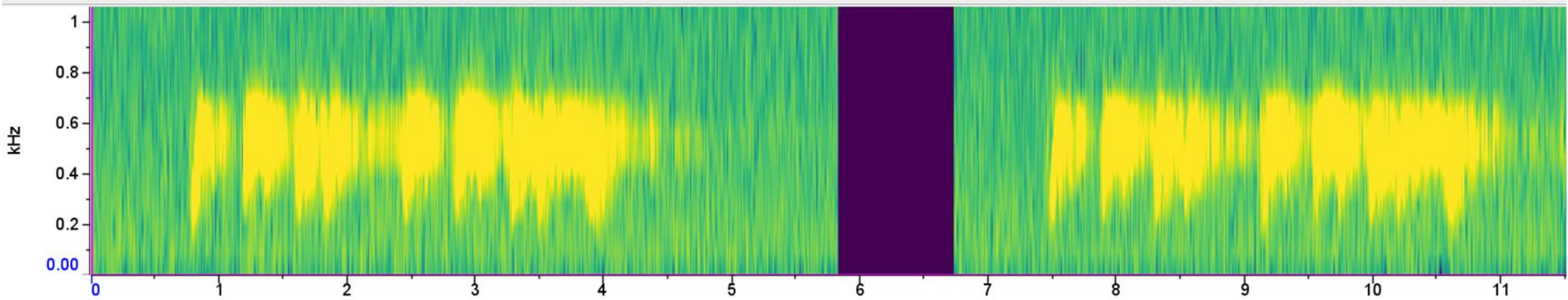
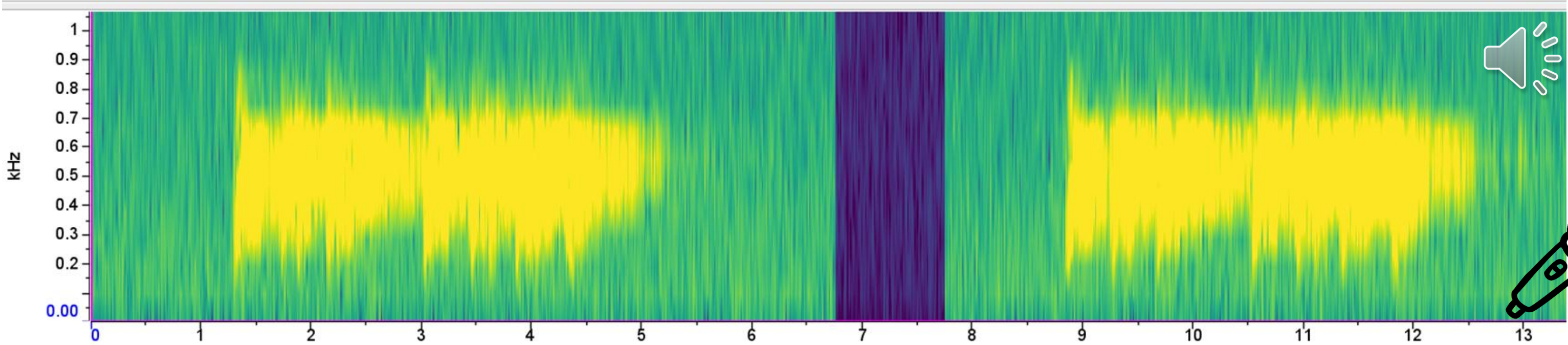


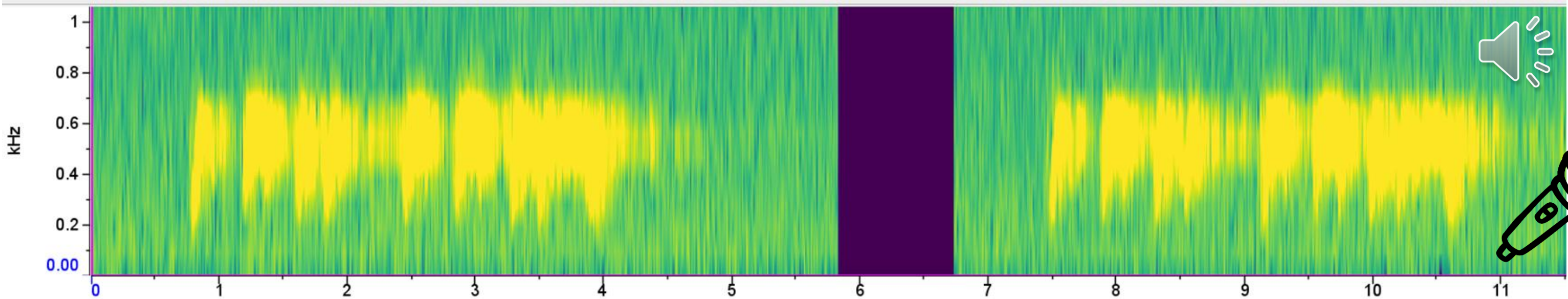
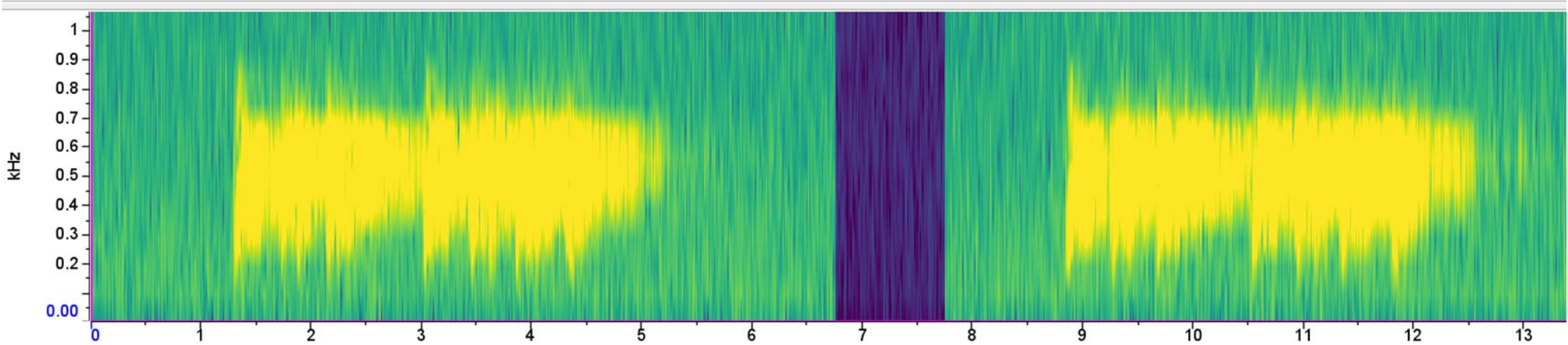
Photo credit: Roy V Rae

- 454 Barred Owl calls
- 10 sites
- Site & owl ID

Can you hear the difference?

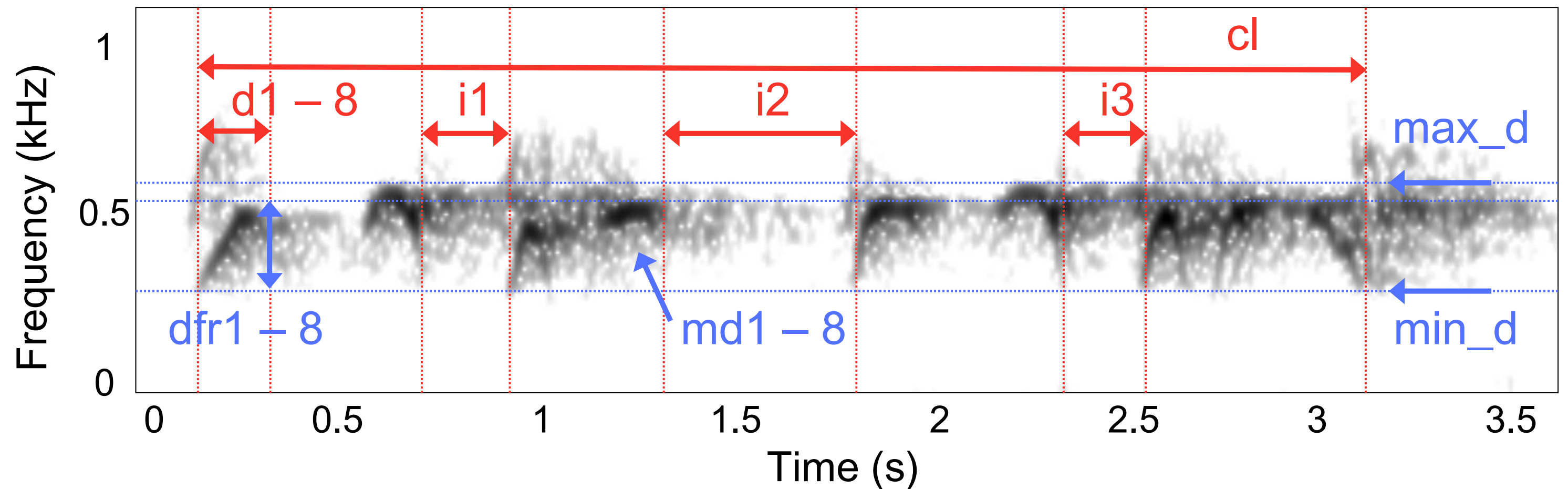
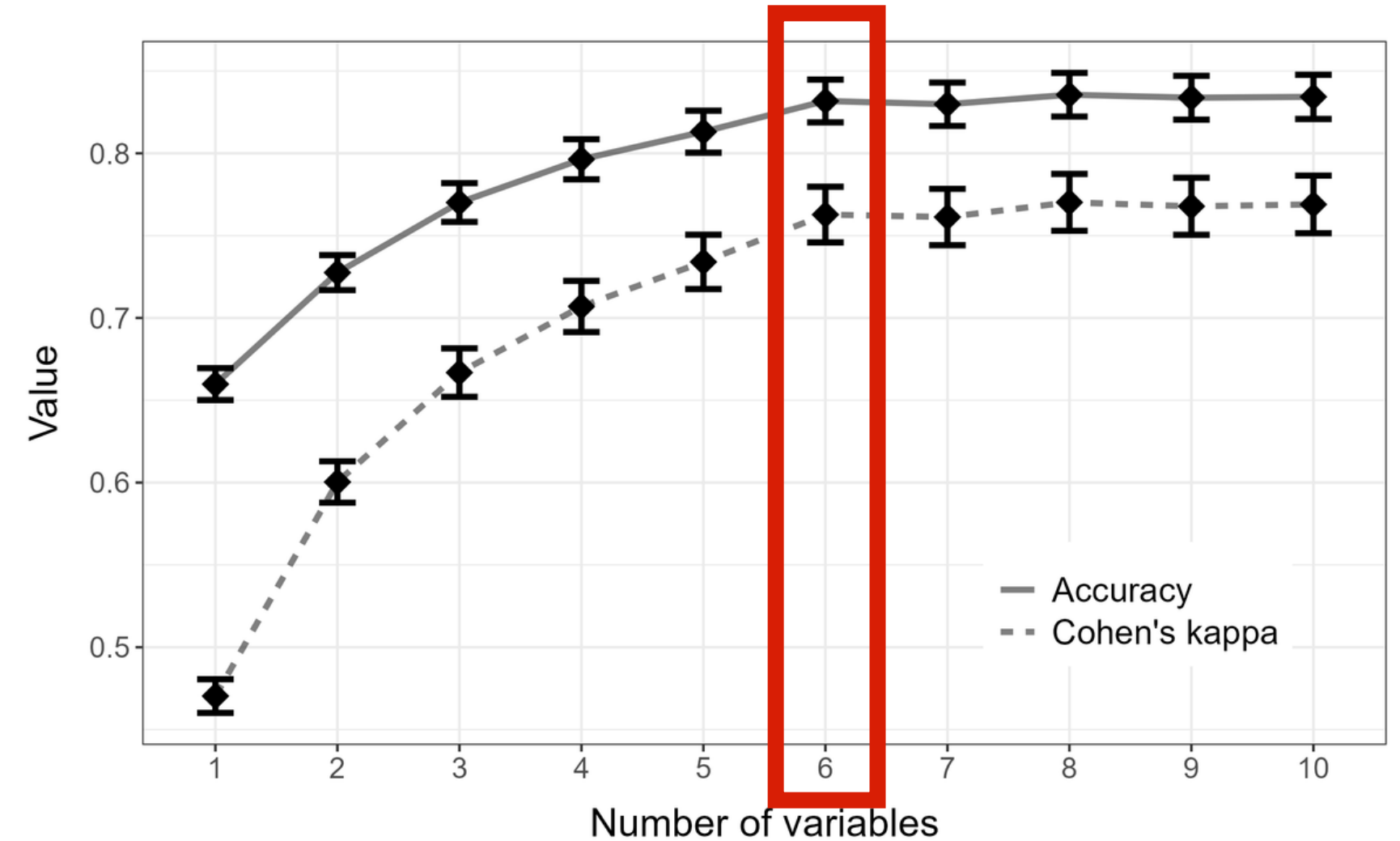


Can you hear the difference?

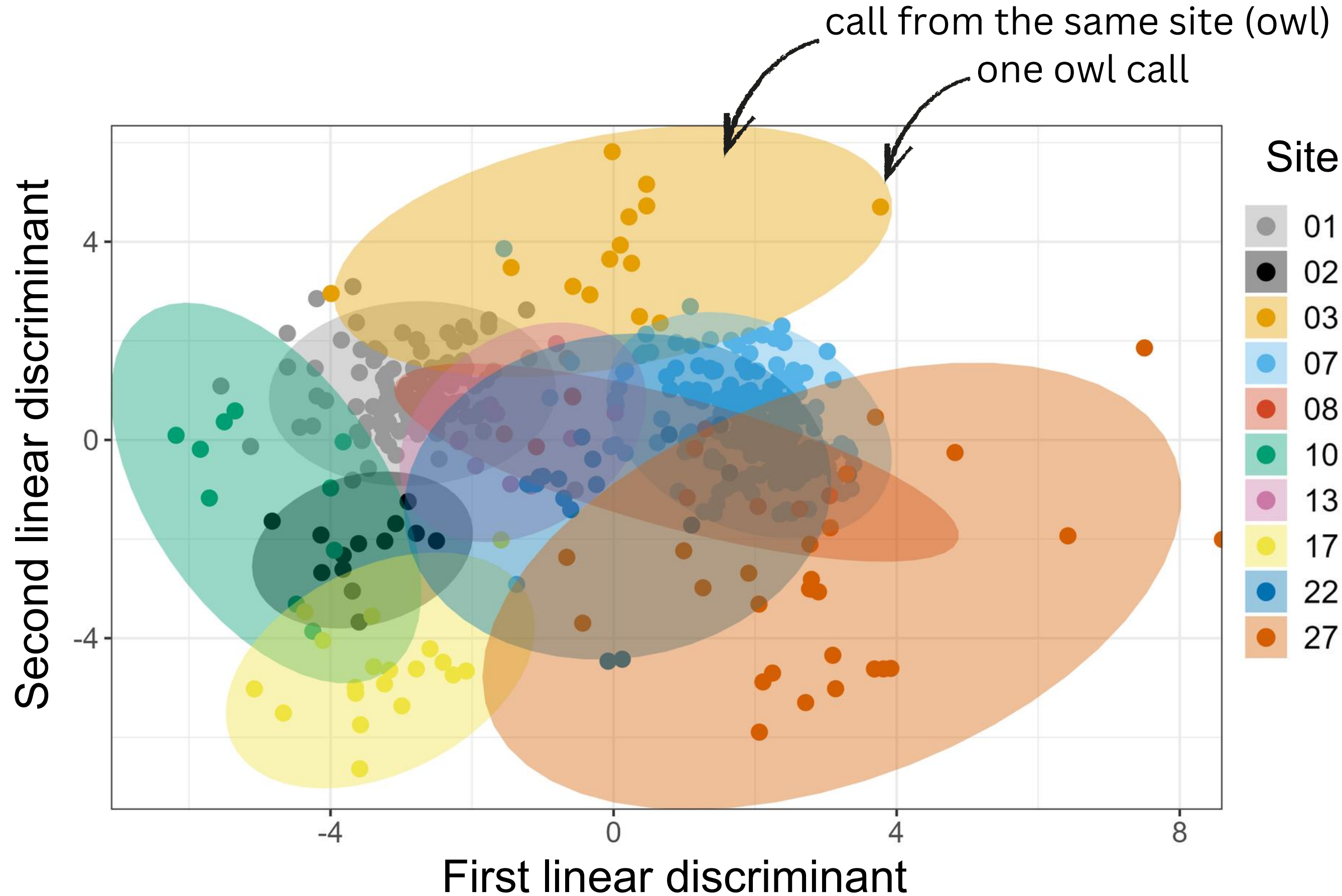


Spectrogram & sound features

- Extract sound features
- **Temporal features (12)**
- **Frequency features (18)**
- Linear discriminant analysis
- Feature selection



Linear discriminant analysis



Performance

- Accuracy = 83%
- Kappa = 76%

Top variables

- **Call length**
- **Interval 4th & 5th**
- **Interval 6th & 7th**
- **Duration of the 8th**
- Frequency of the 6th
- Duration of the 3rd note

Individual identification using passive acoustic monitoring

1. Linear discriminant analysis → 83% accuracy
- 2. Temporal features** provided strong discrimination of calls
3. Future Barred Owl monitoring



Barred Owl calls, as captured by passive acoustic monitoring, exhibit sufficient feature variation to identify individuals.

Machine learning output → ecological insight

Within species



**Identifying
individual owls**

Single species



**Finding spatiotemporal
vocal activity patterns**

Multiple species

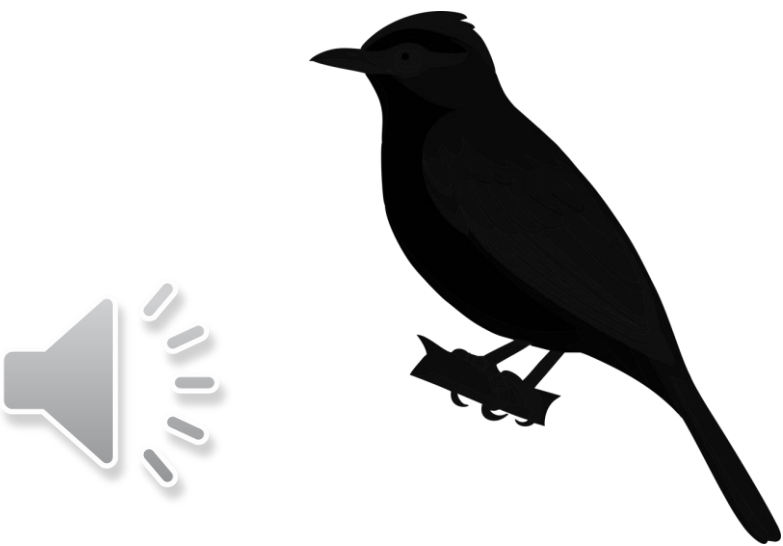
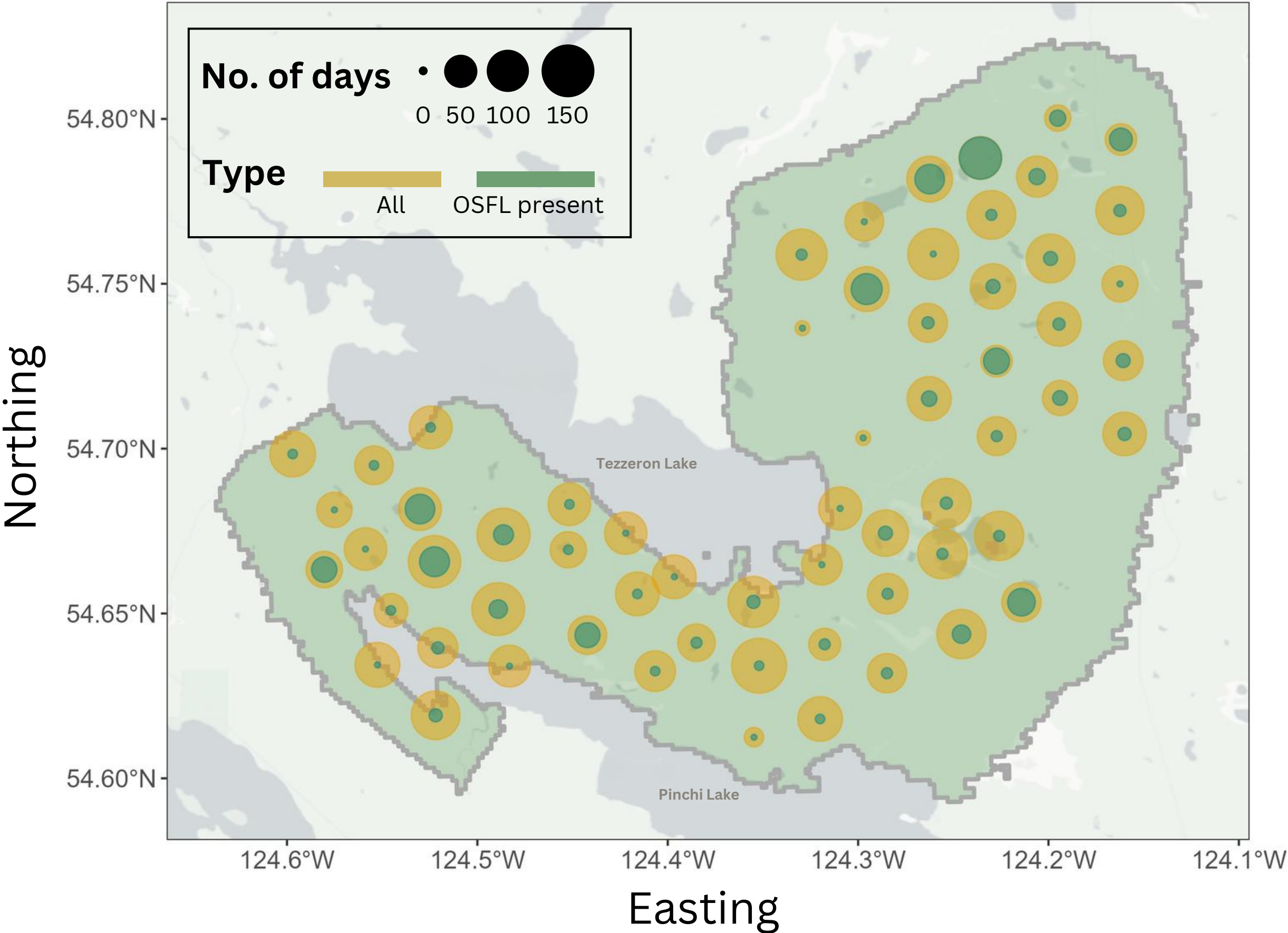


**Linking habitat
features to richness**



Identifying the seasonal activity pattern and the habitat use with passive acoustic monitoring

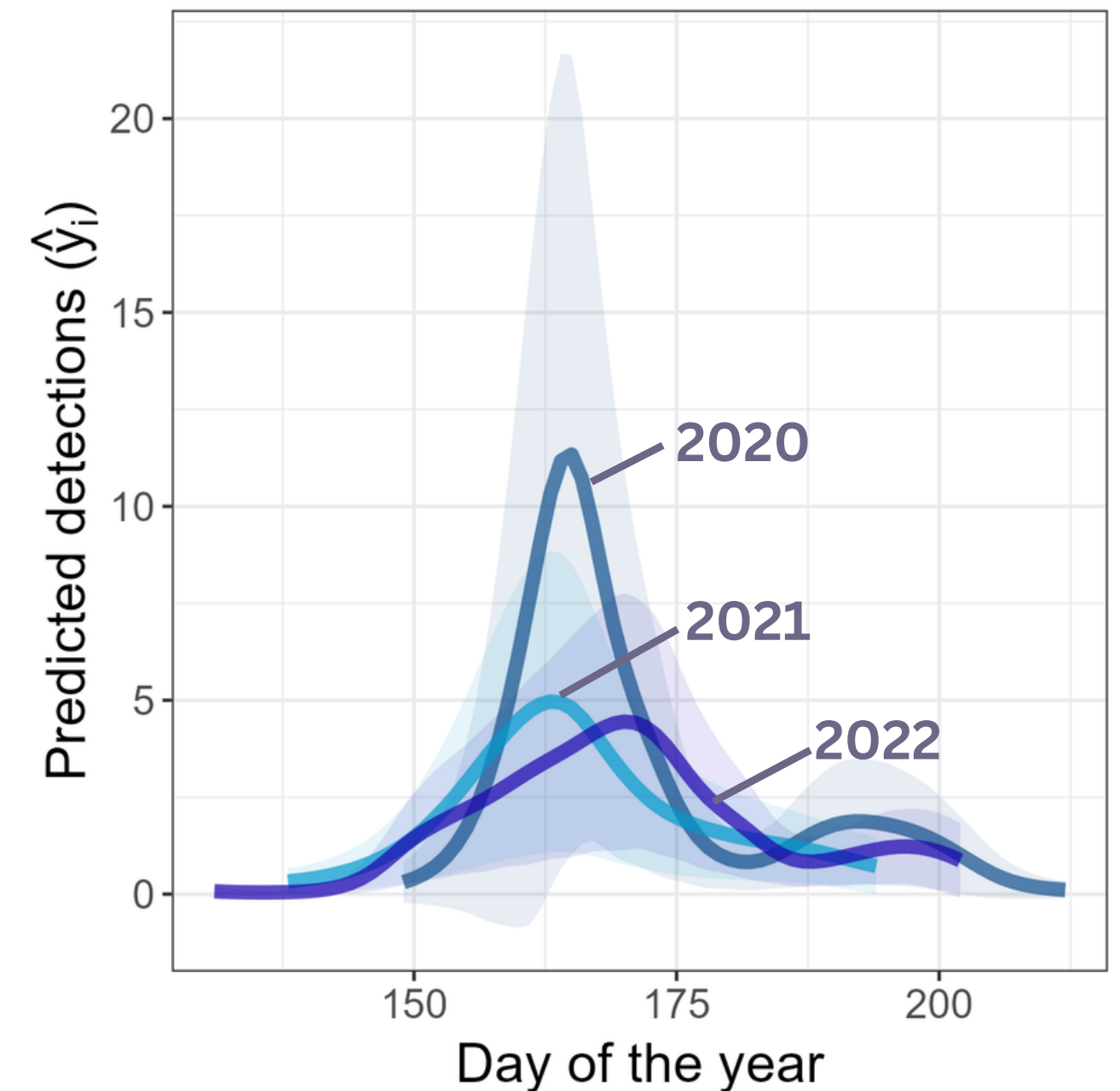
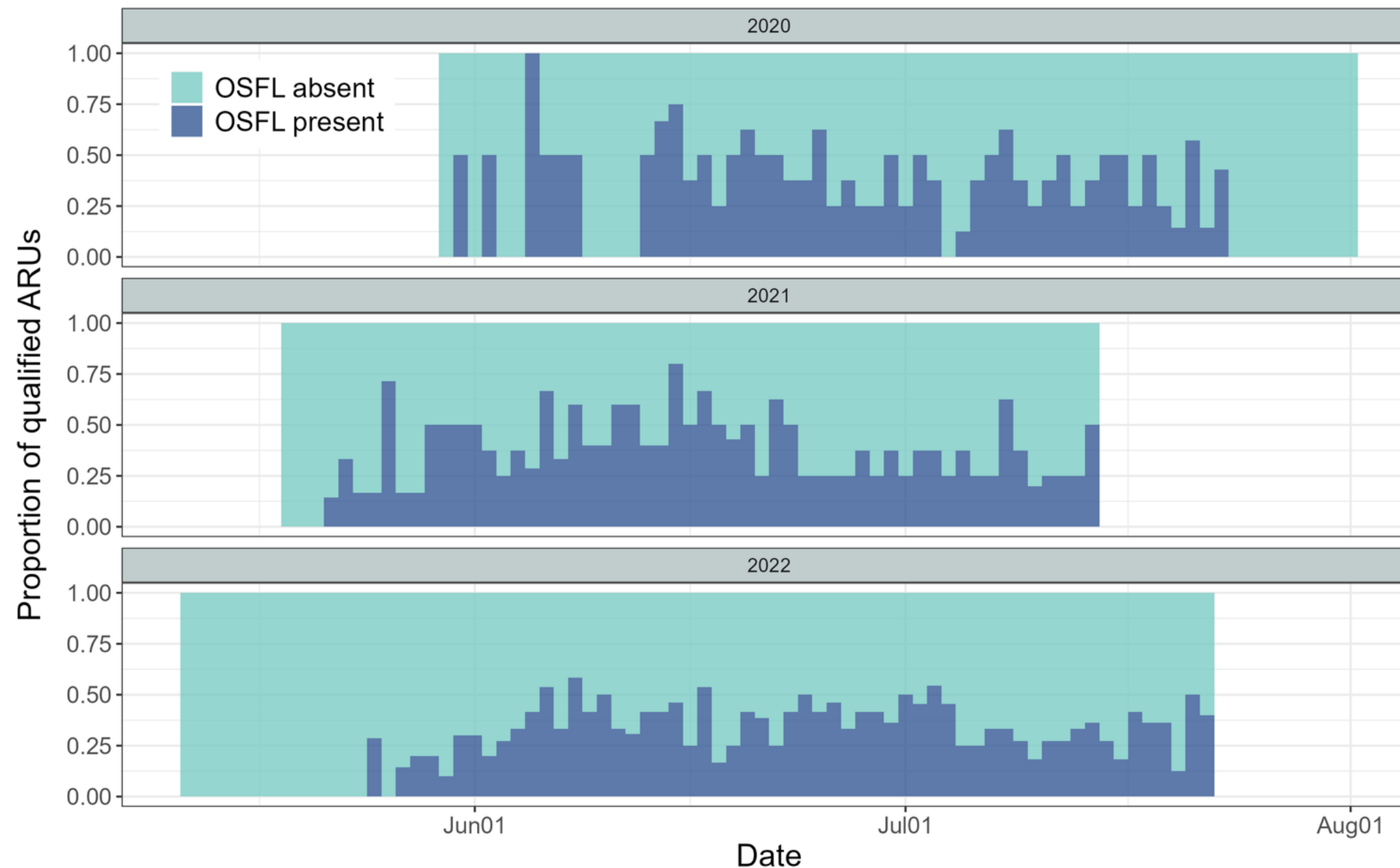
Olive-sided Flycatcher



- “Quick-three beers!”
- Northeast part

Seasonal activity pattern

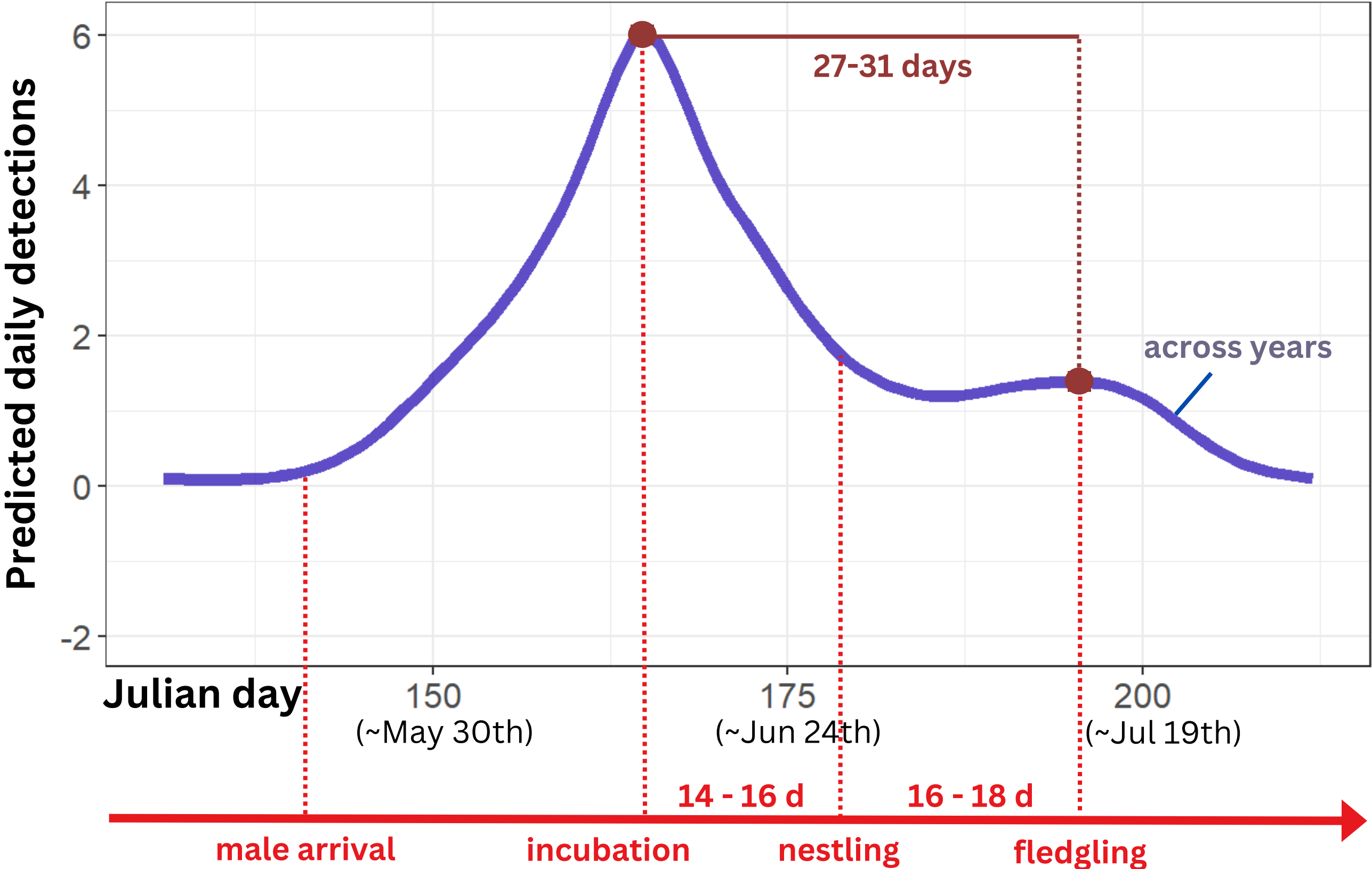
- Activities centralized in June / July
- Interannual variation in peak timing and magnitude
- Primary peak (mid-June) and secondary peak (mid-July)



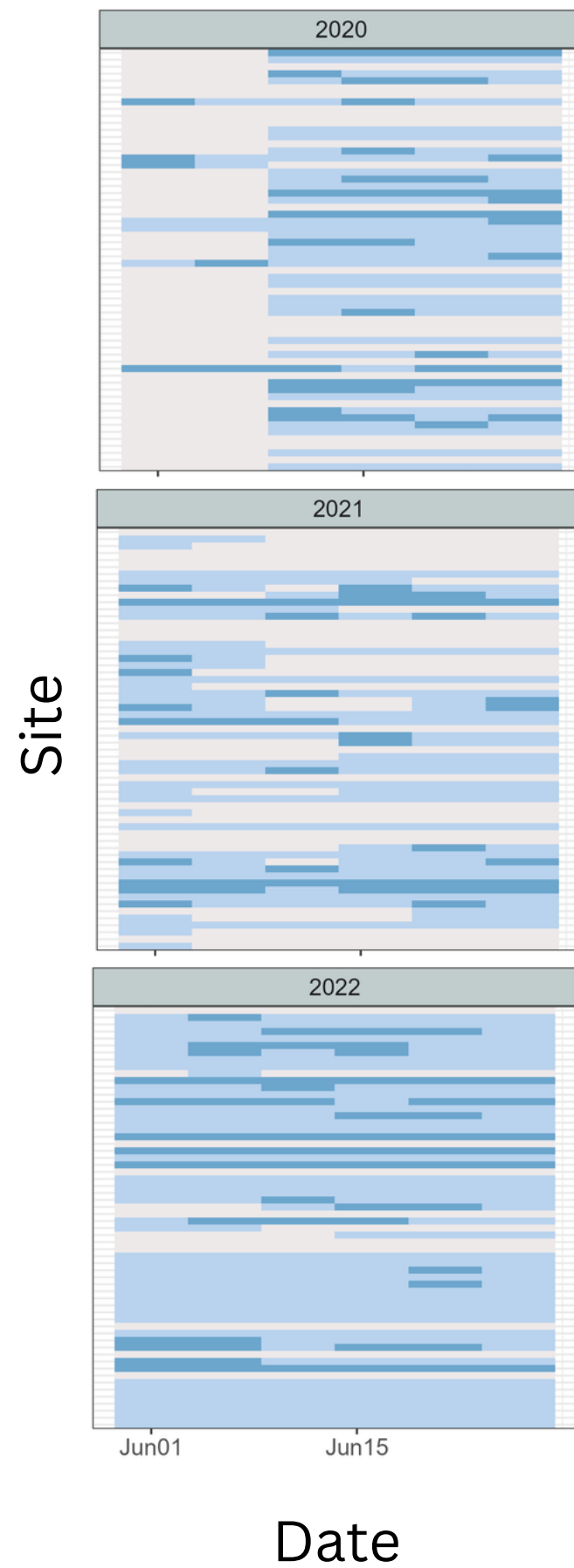
Linking vocal activity to breeding phenology

Vocal activity

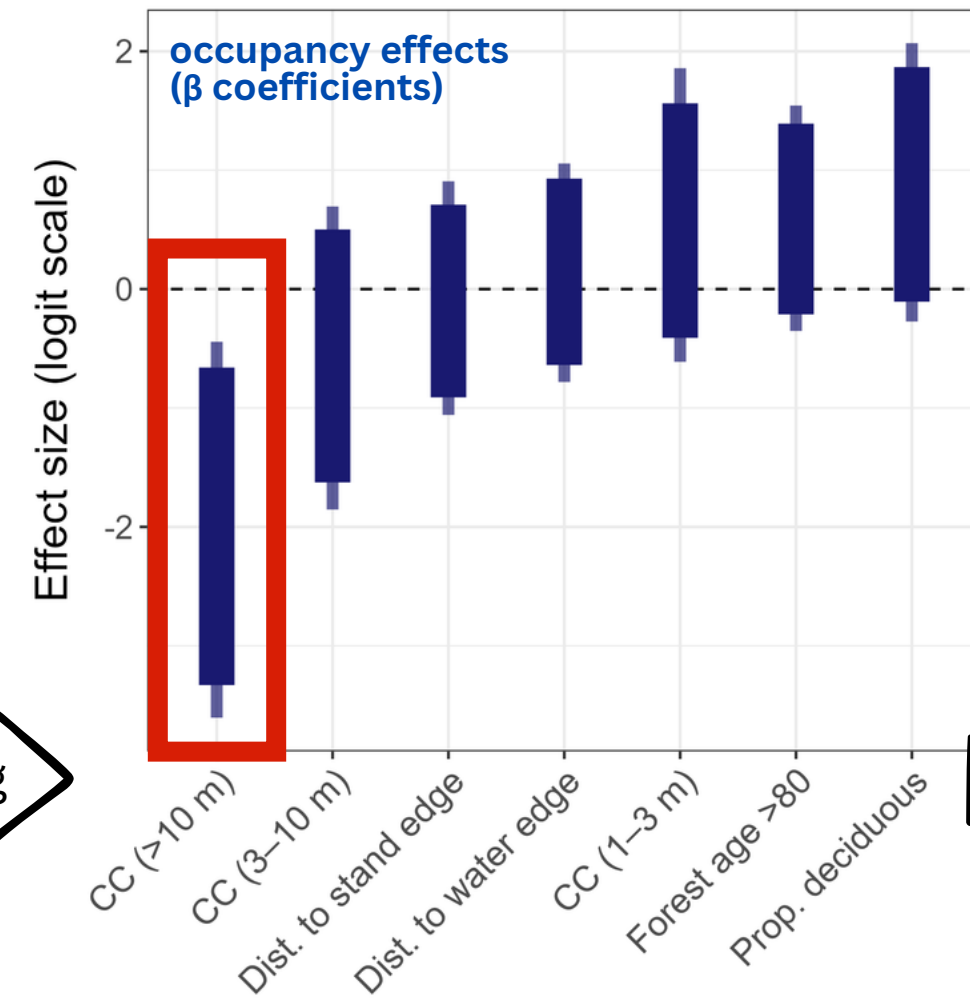
Breeding phenology



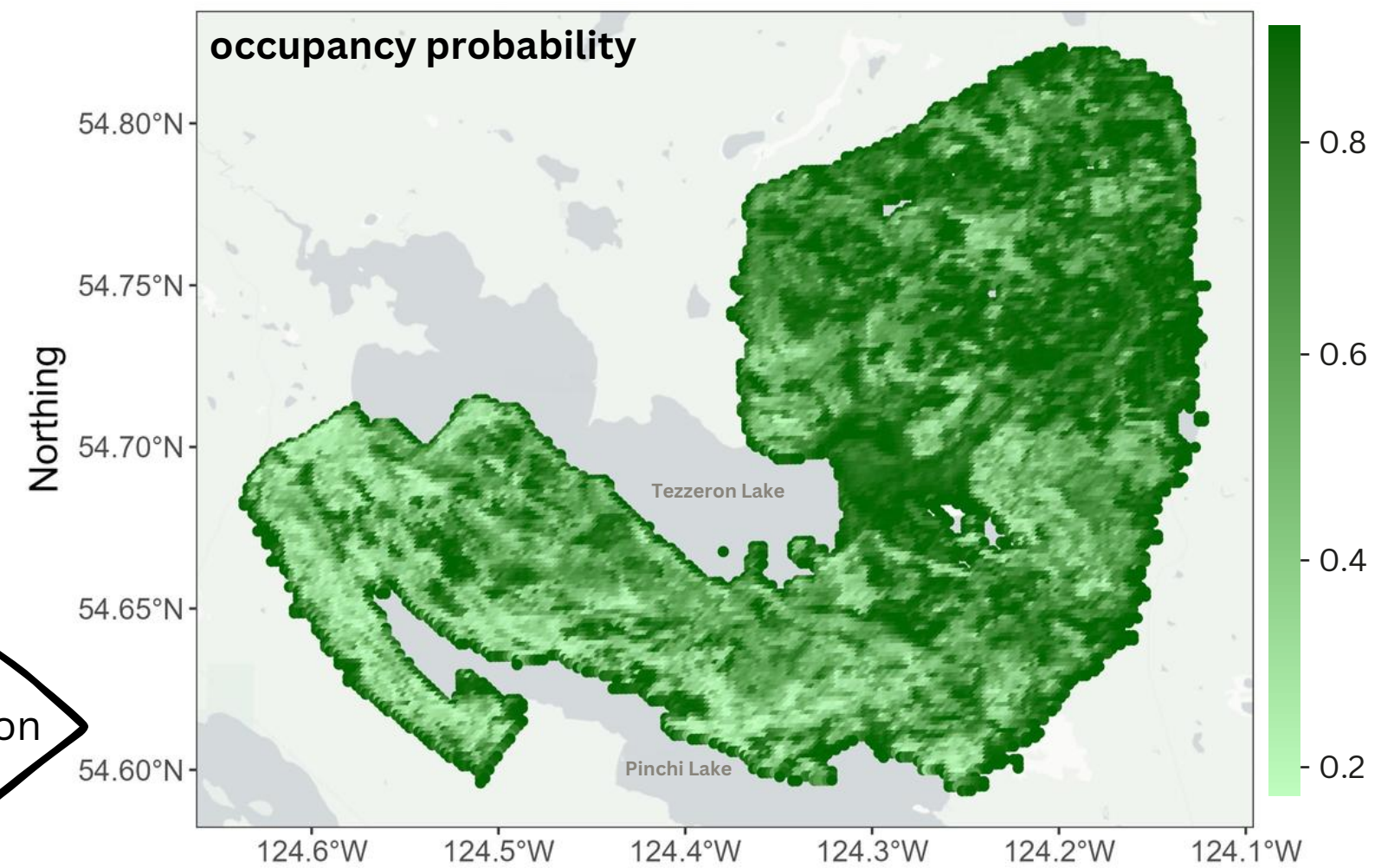
Habitat use



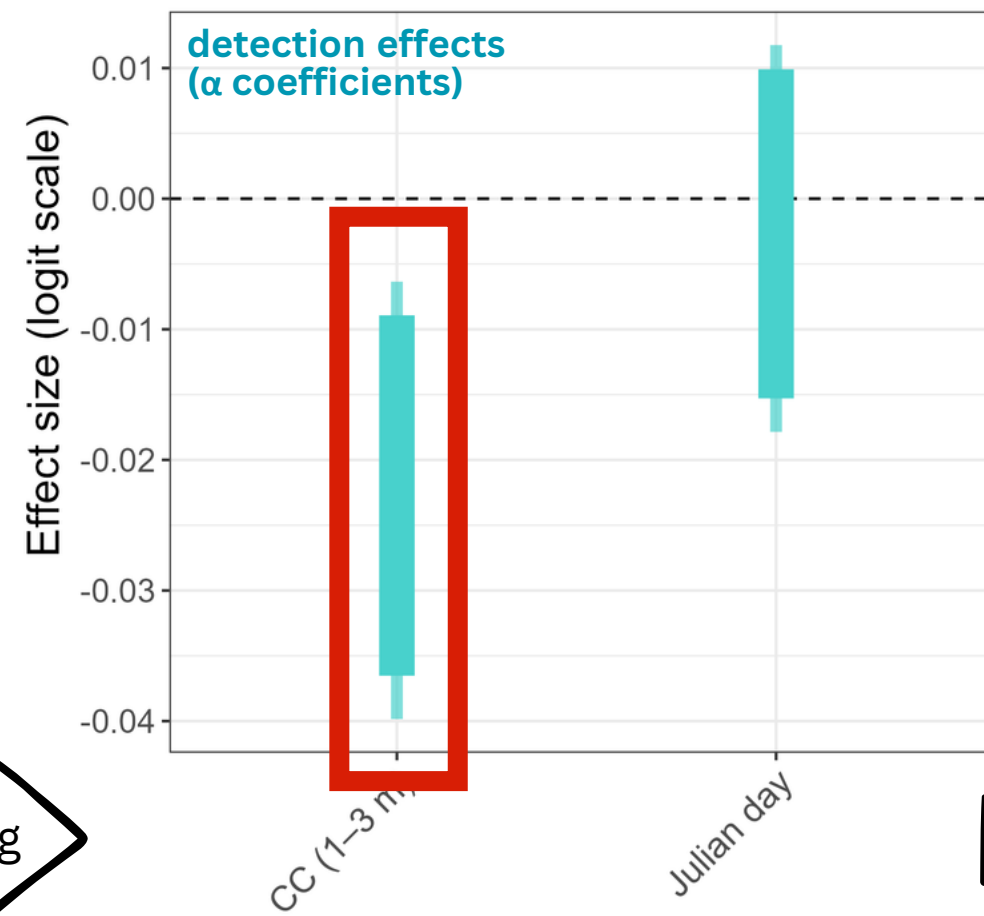
Modelling



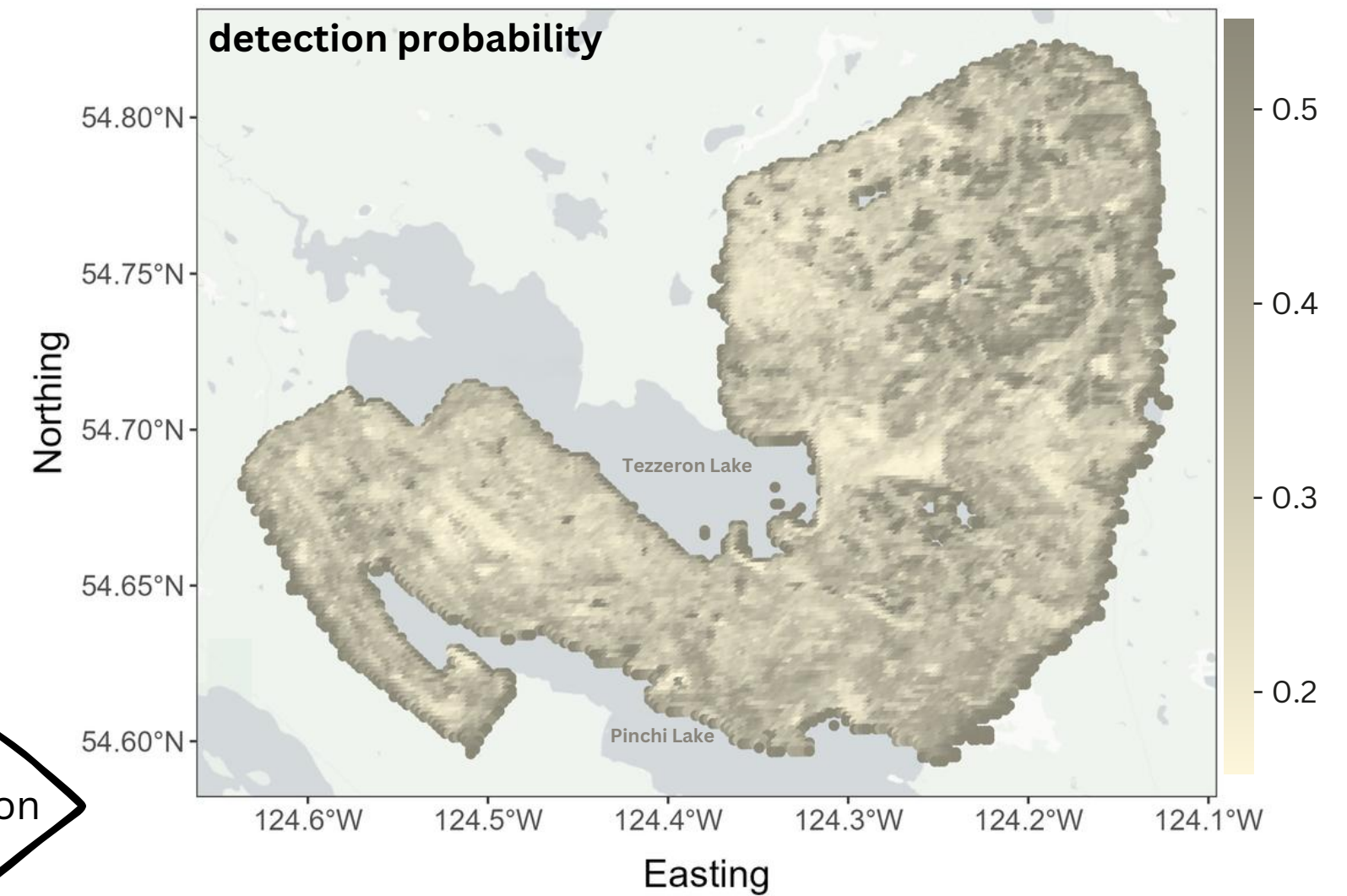
Prediction



Modelling



Prediction



Spatiotemporal vocal activity patterns

1. Activity peaks at **mid-June**, with a **second peak in mid-July**
2. Vocal activity linked with breeding phenology
3. Prefer sites with **sparse upper-crown density**; detectability ↓
with ↑ lower-crown density



Passive acoustic monitoring can reveal species-specific vocal activity patterns, enhancing the understanding of behaviour.

Machine learning output → ecological insight

Within species



**Identifying
individual owls**

Single species



**Finding spatiotemporal
vocal activity patterns**

Multiple species



**Linking habitat
features to richness**

Machine learning output → ecological insight



Within species

Single species

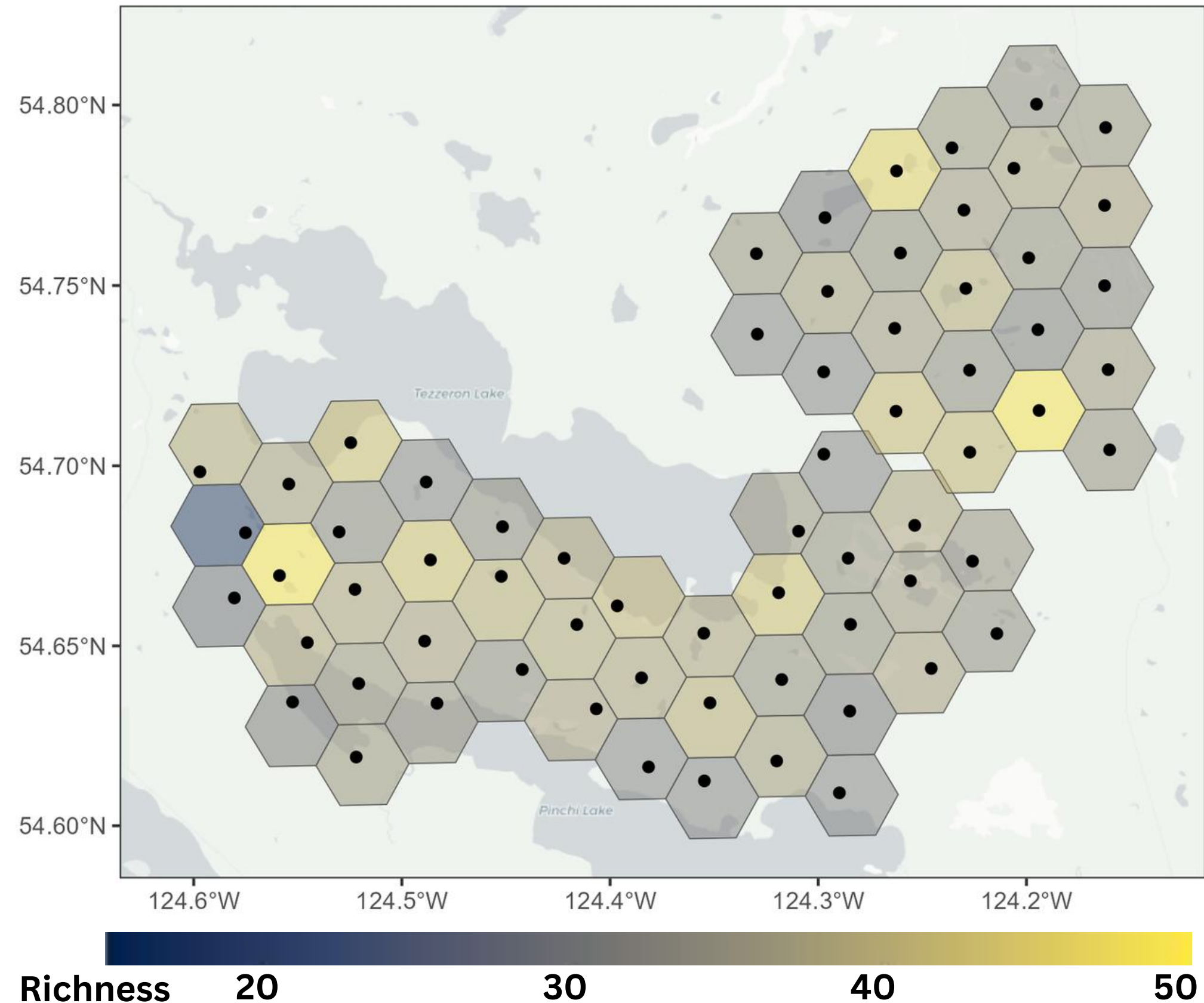
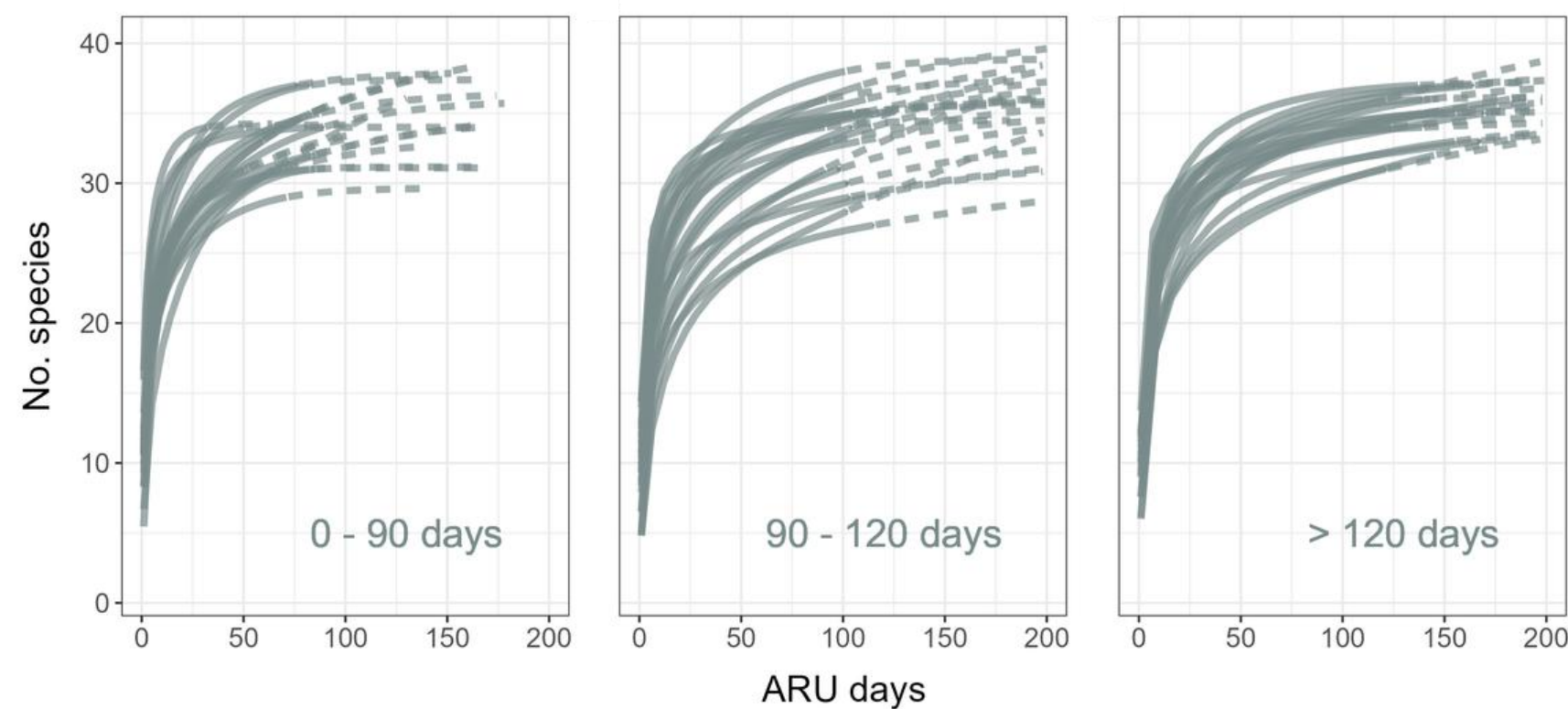


Multiple species

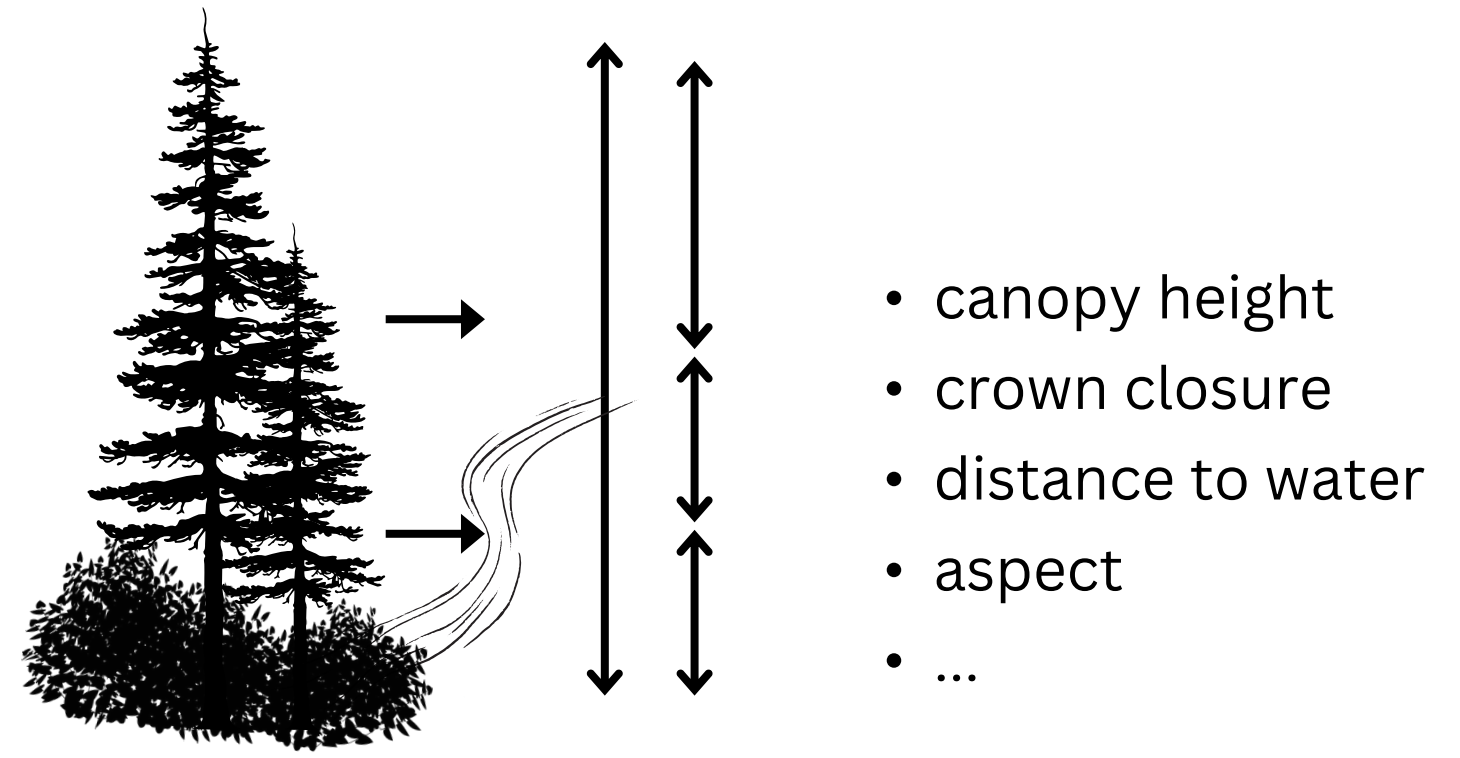
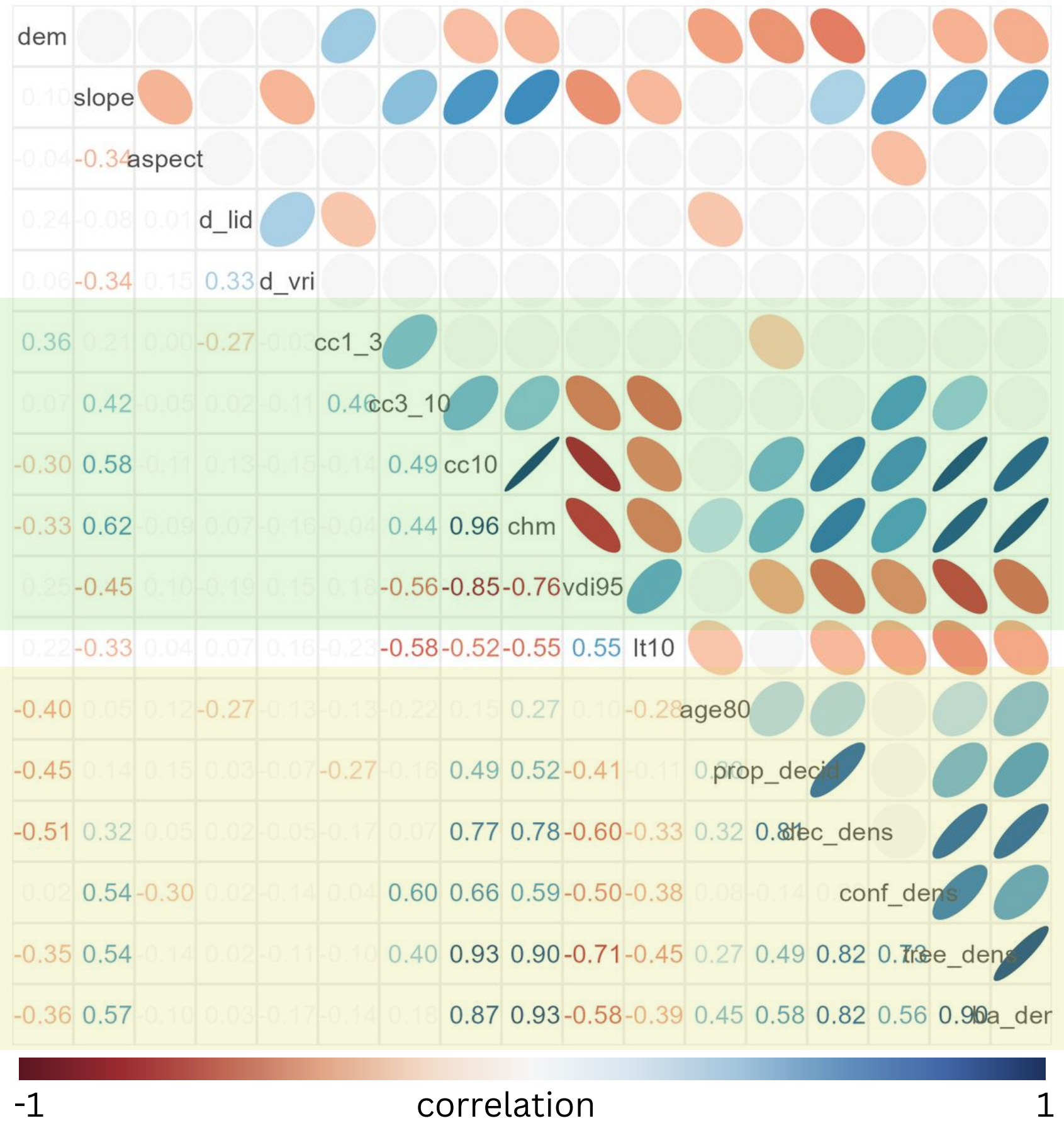
Quantifying breeding songbird richness across the research forest and linking it to the LiDAR habitat structure

Richness distribution across landscape

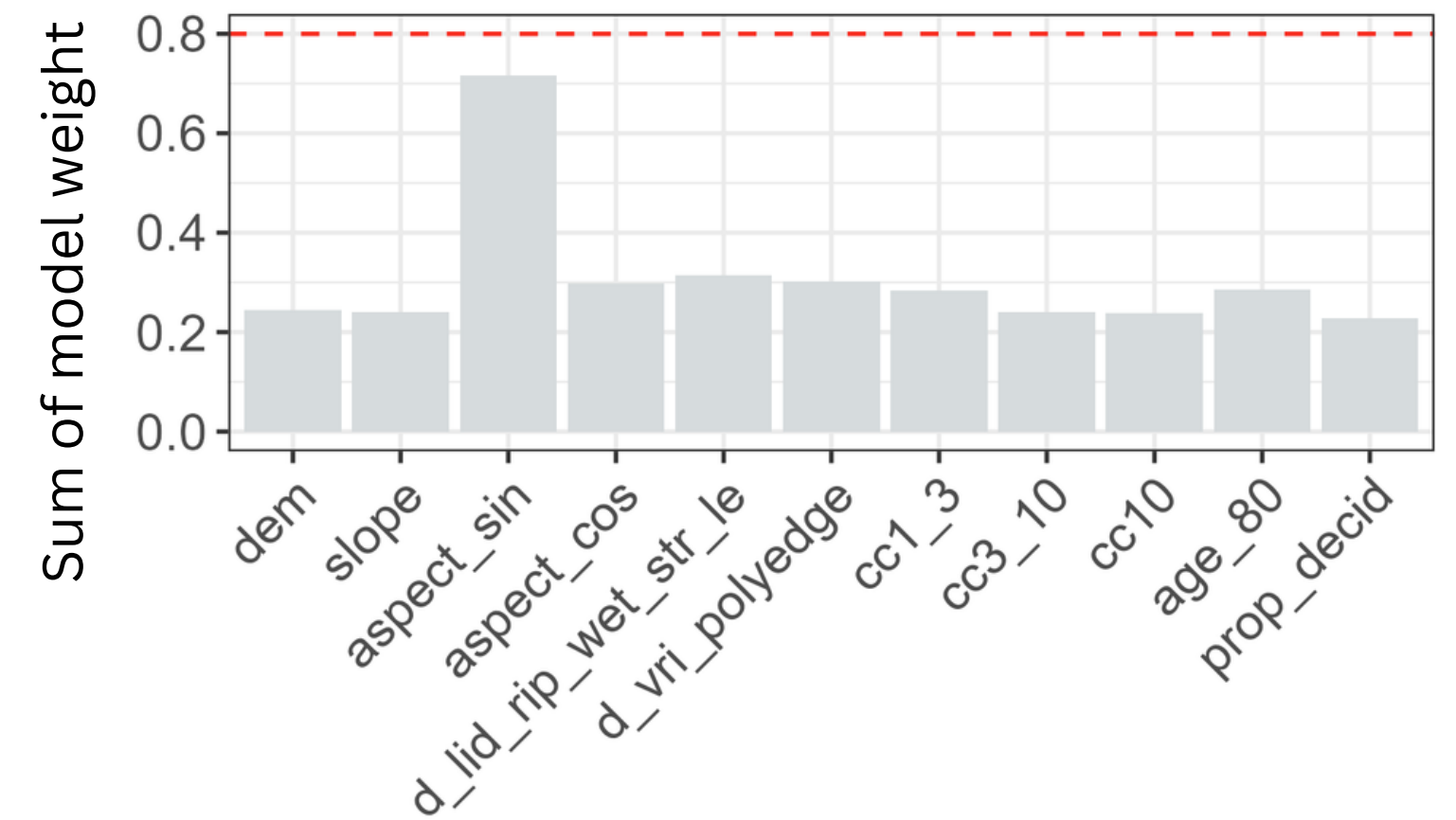
- Breeding songbirds (19 - 38 sp.)
- Unbalanced effort
- Asymptotic richness



LiDAR covariates



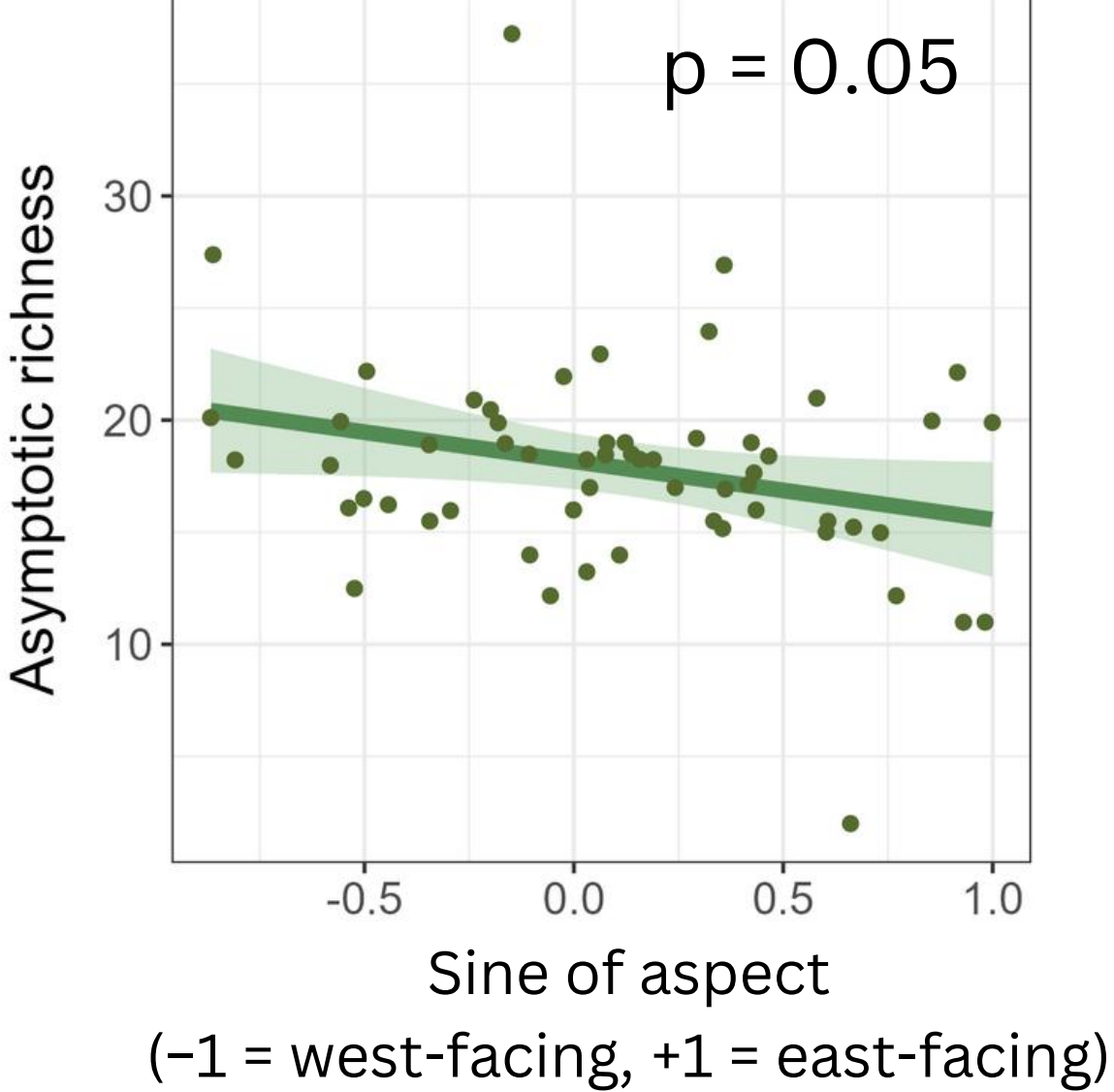
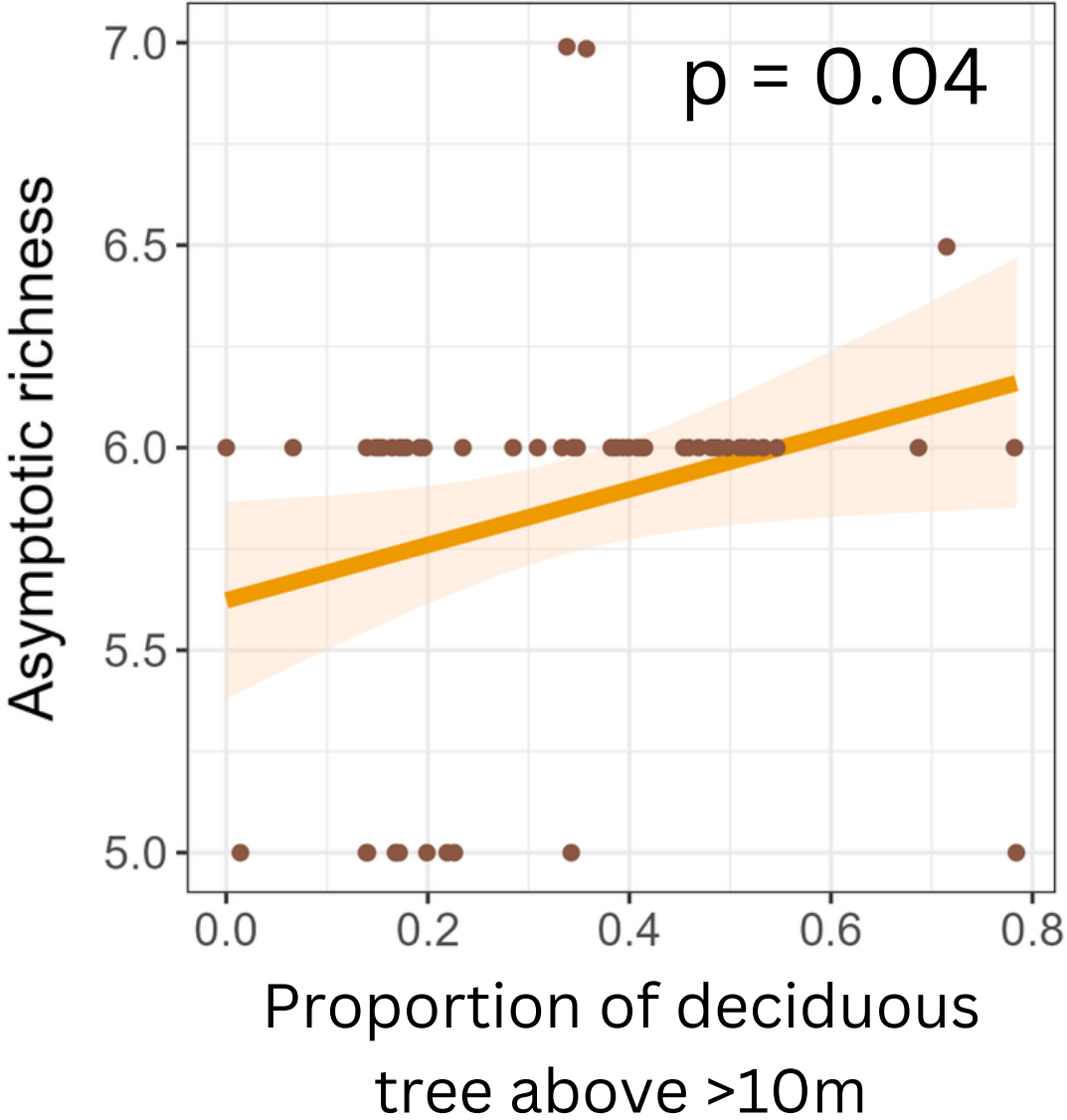
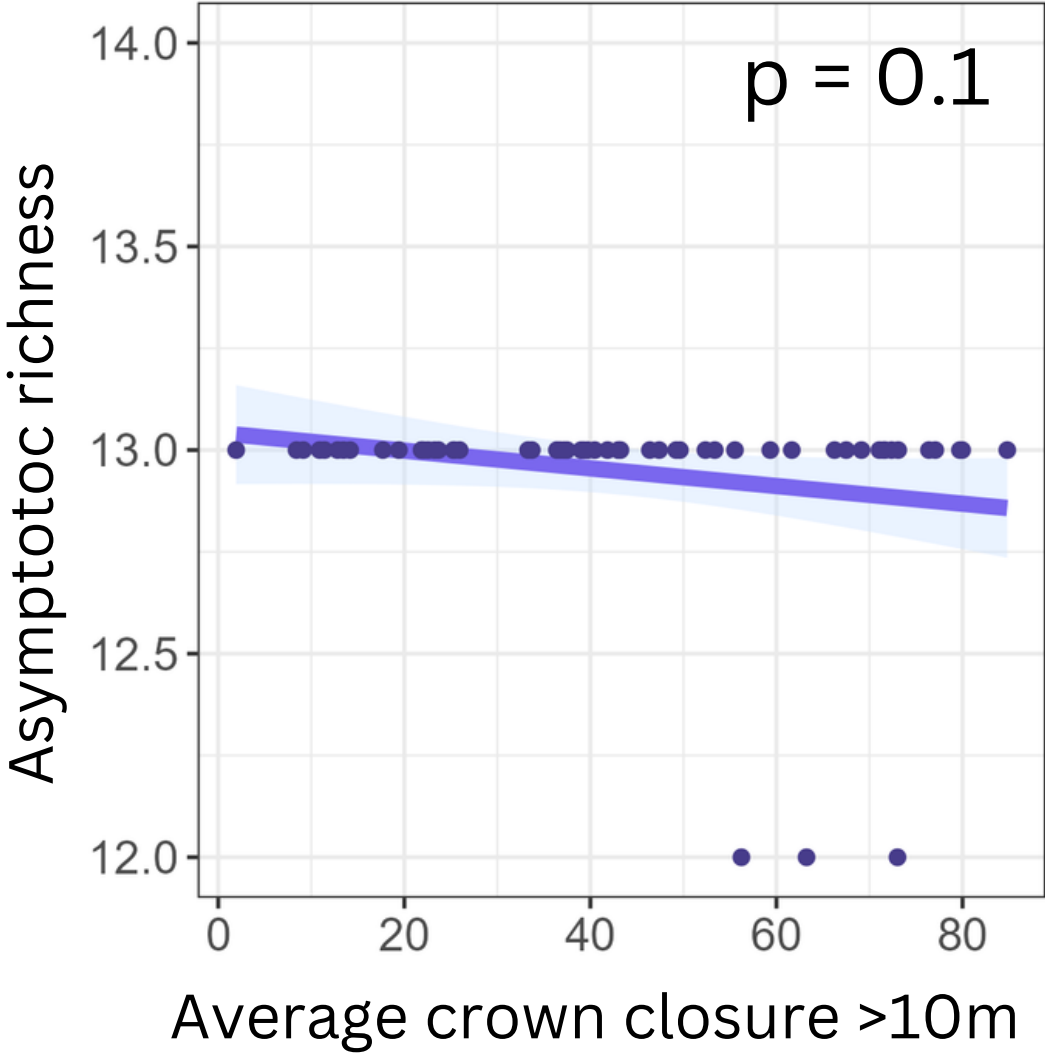
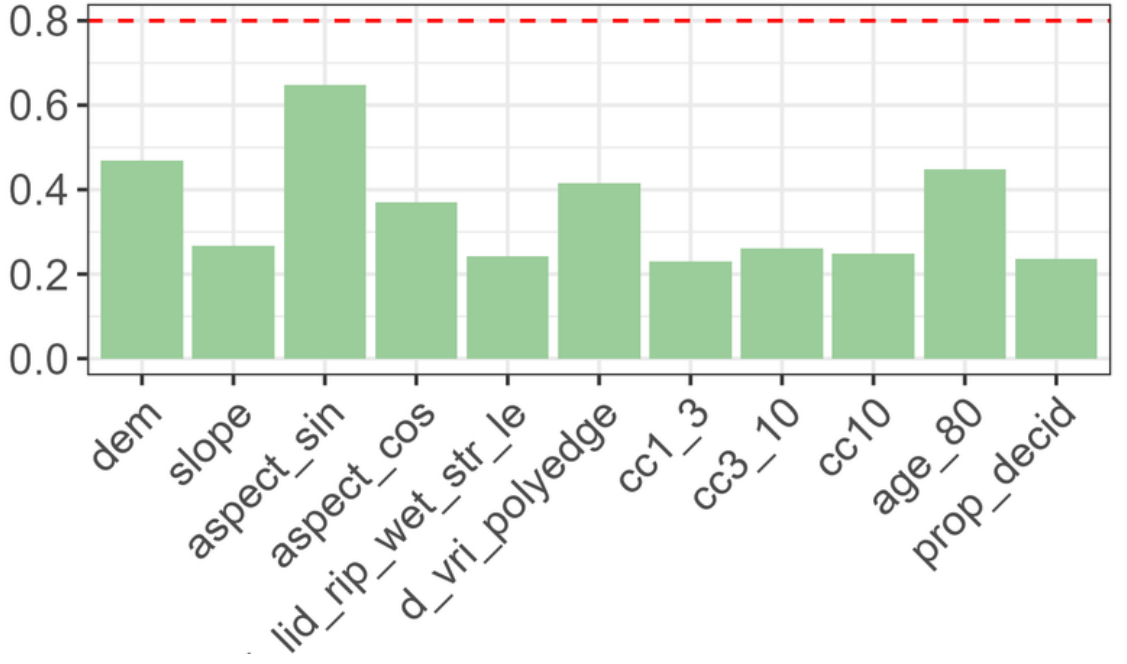
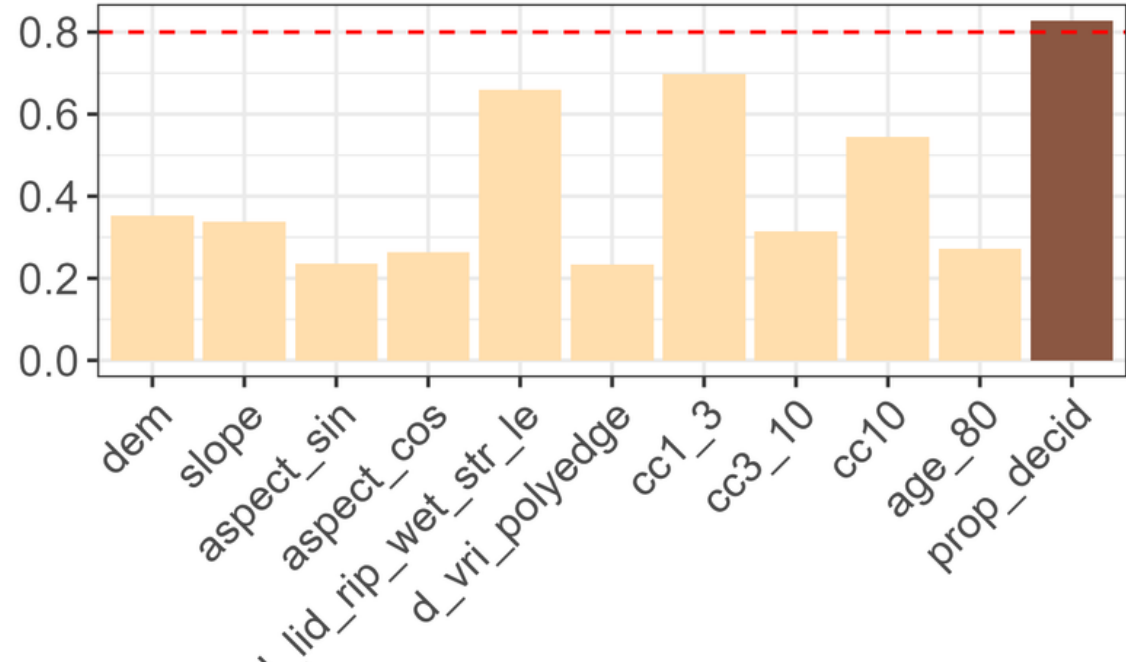
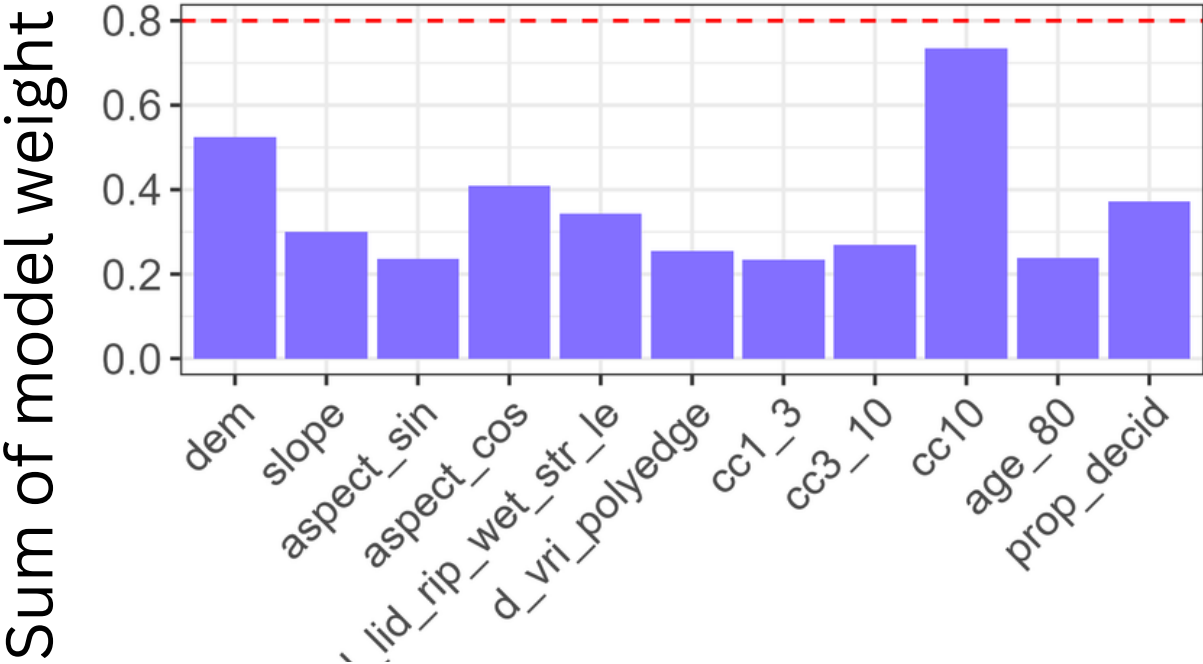
- Variable reduction (corr. & VIF)
- No significant covariate



Group species by their occurrence similarity

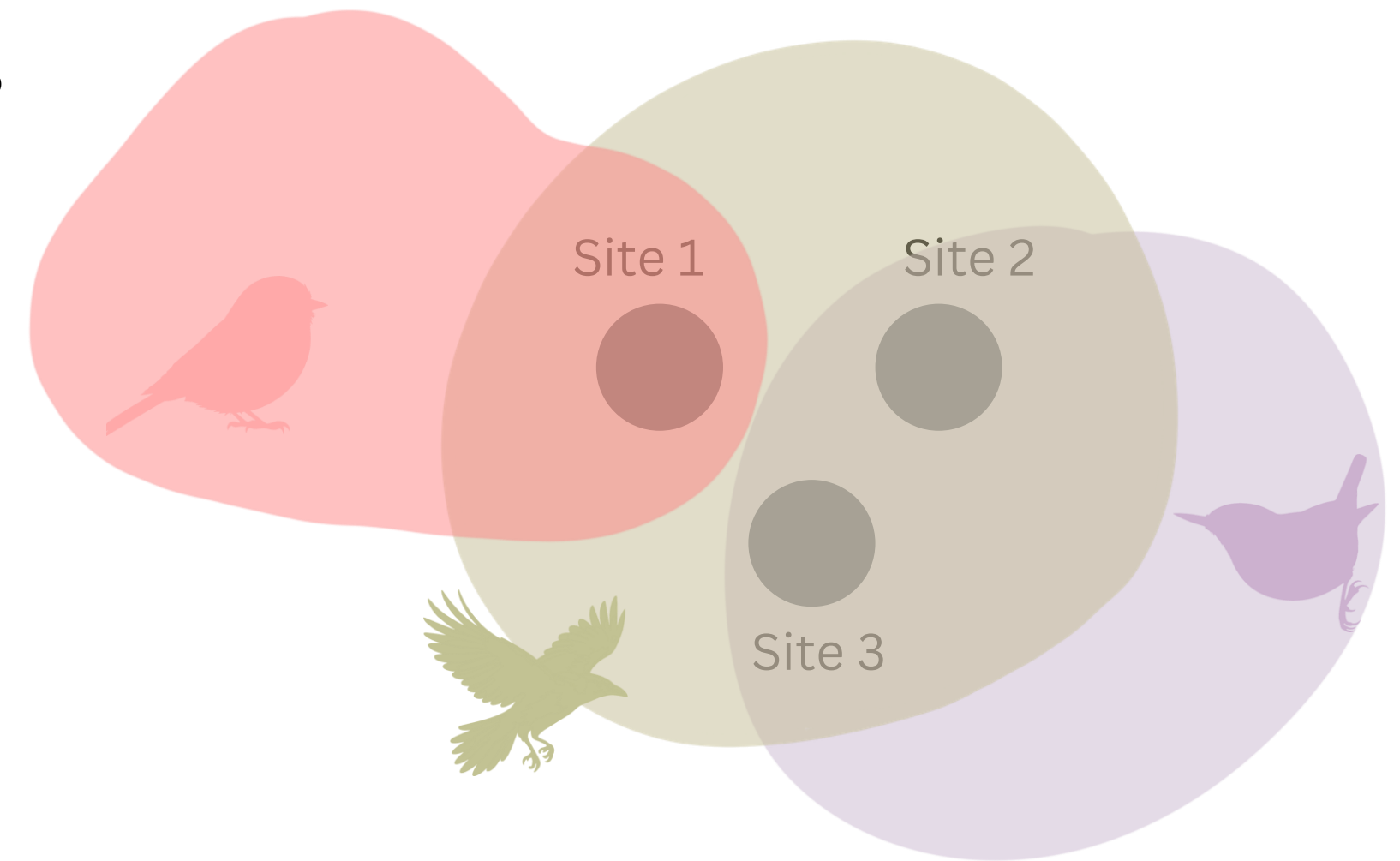


Richness relationship to LiDAR covariates by groups



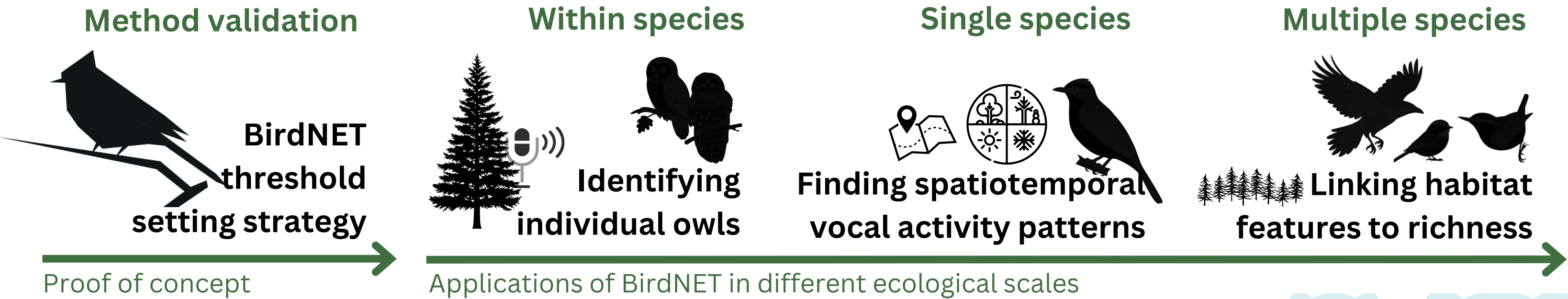
Linking richness distribution with habitat features

1. Breeding songbirds' **richness across the research forest**
2. **Diverse habitat use** of bird species
3. Relatively uniform richness across sites

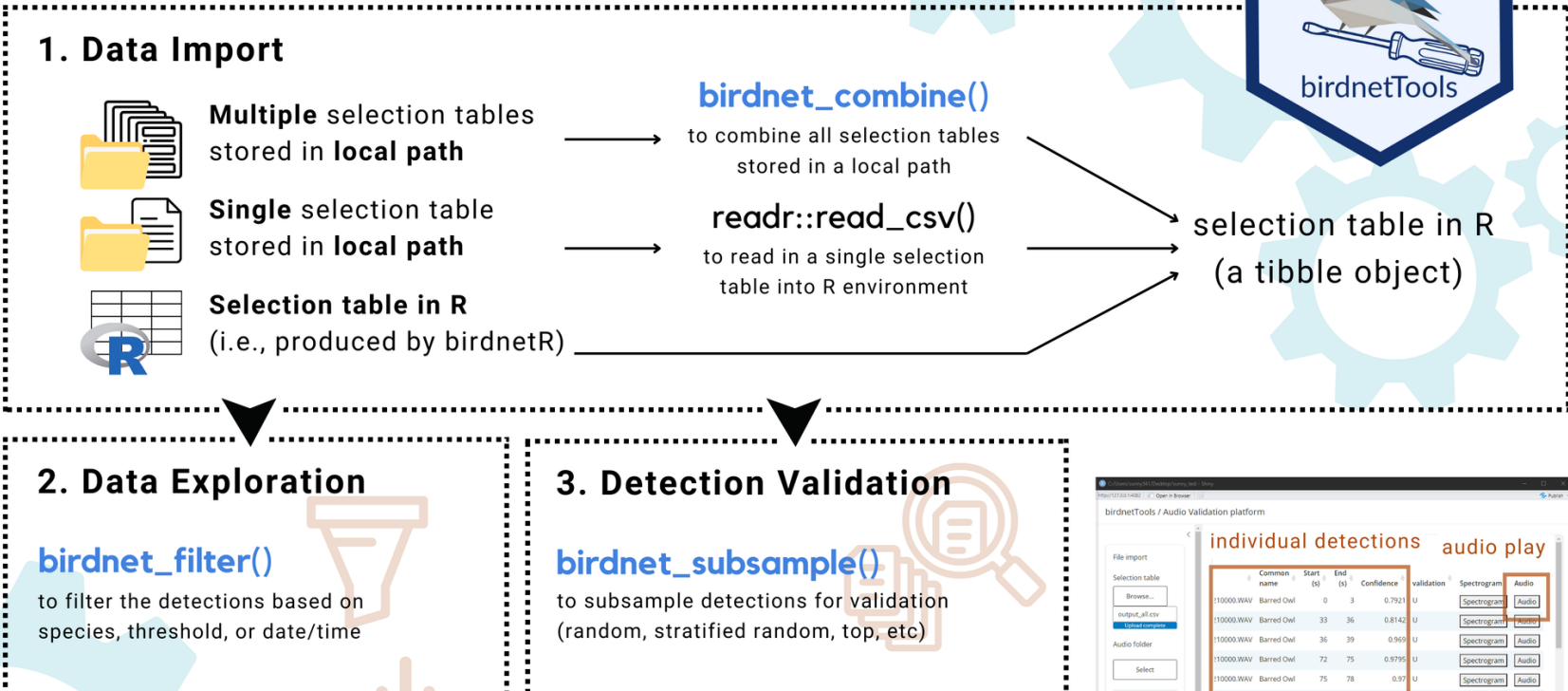


Passive acoustic monitoring with machine learning can be an effective tool in quantifying avian richness in a landscape.

This thesis as a whole



birdnetTools::work flow



Tseng and Kahl (under review) Journal of Open Source Software



Thank you!

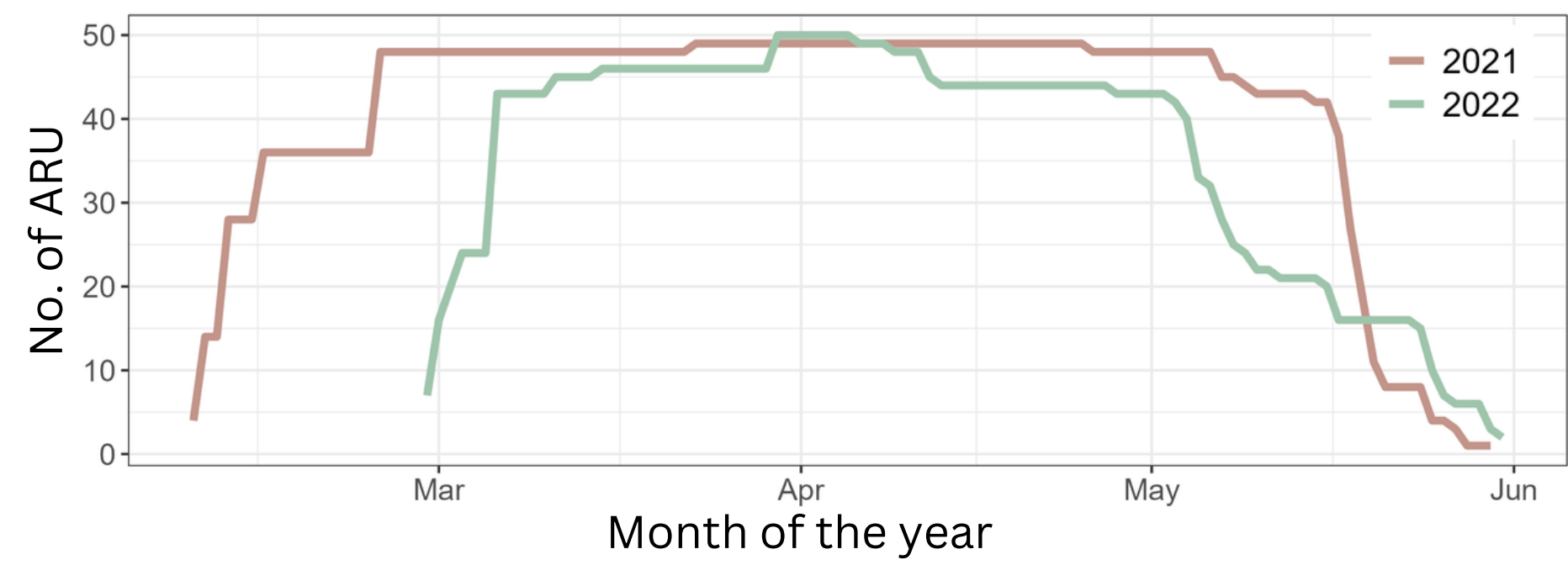
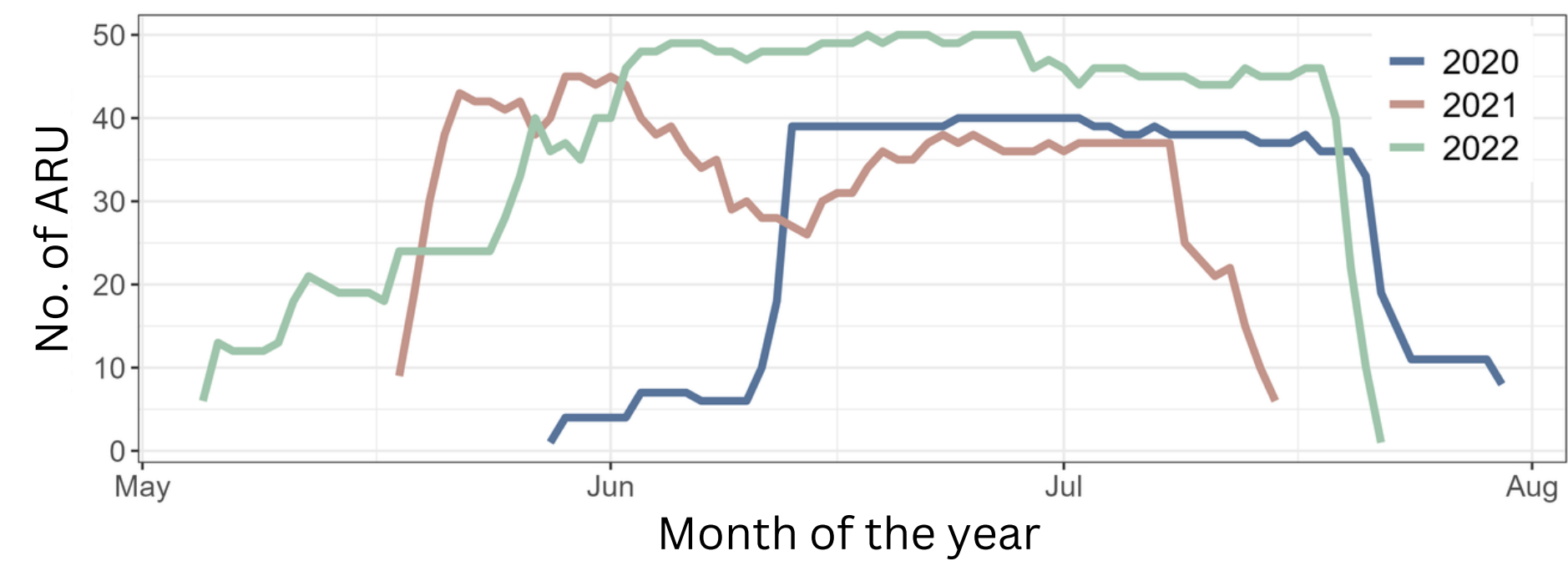
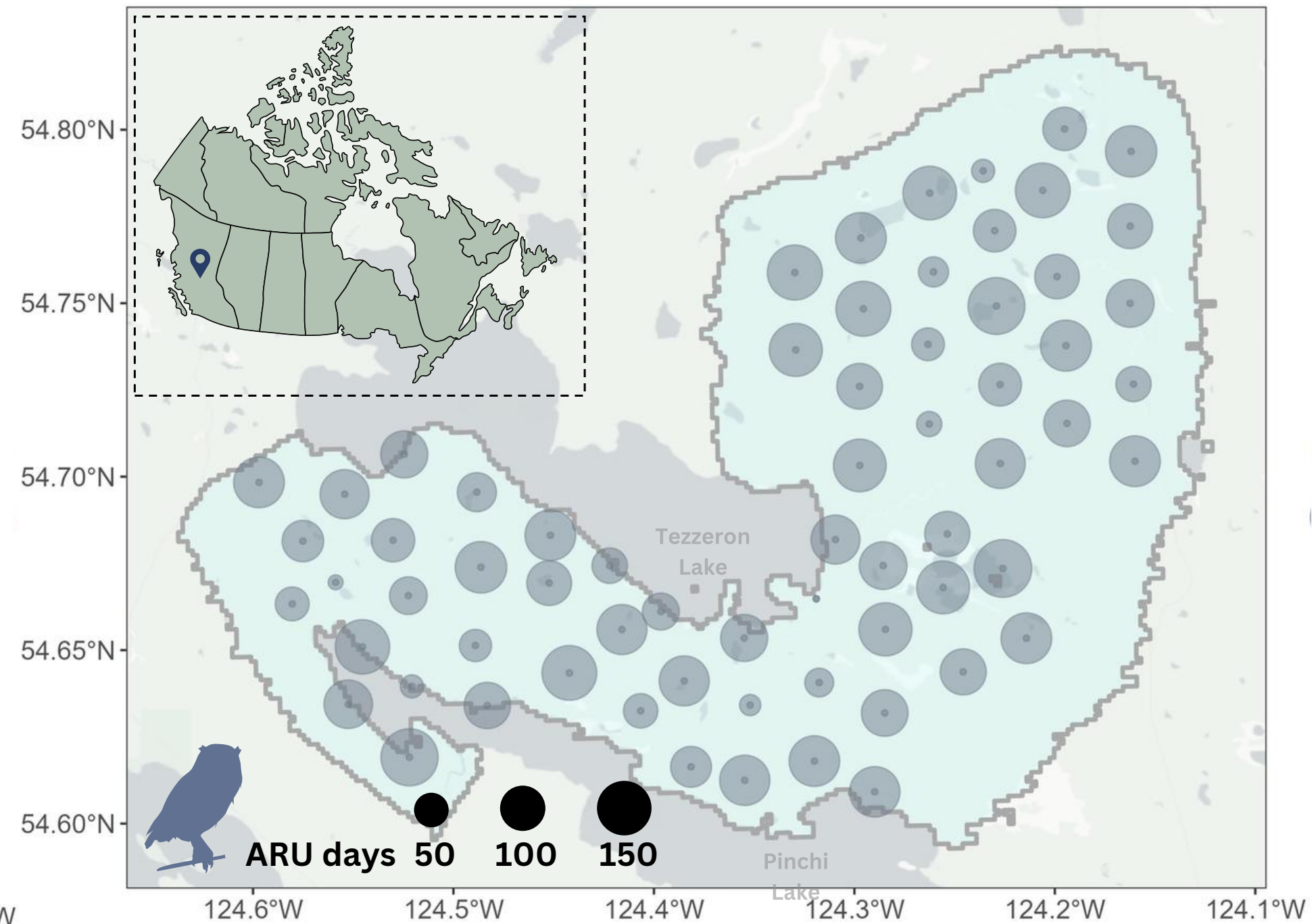
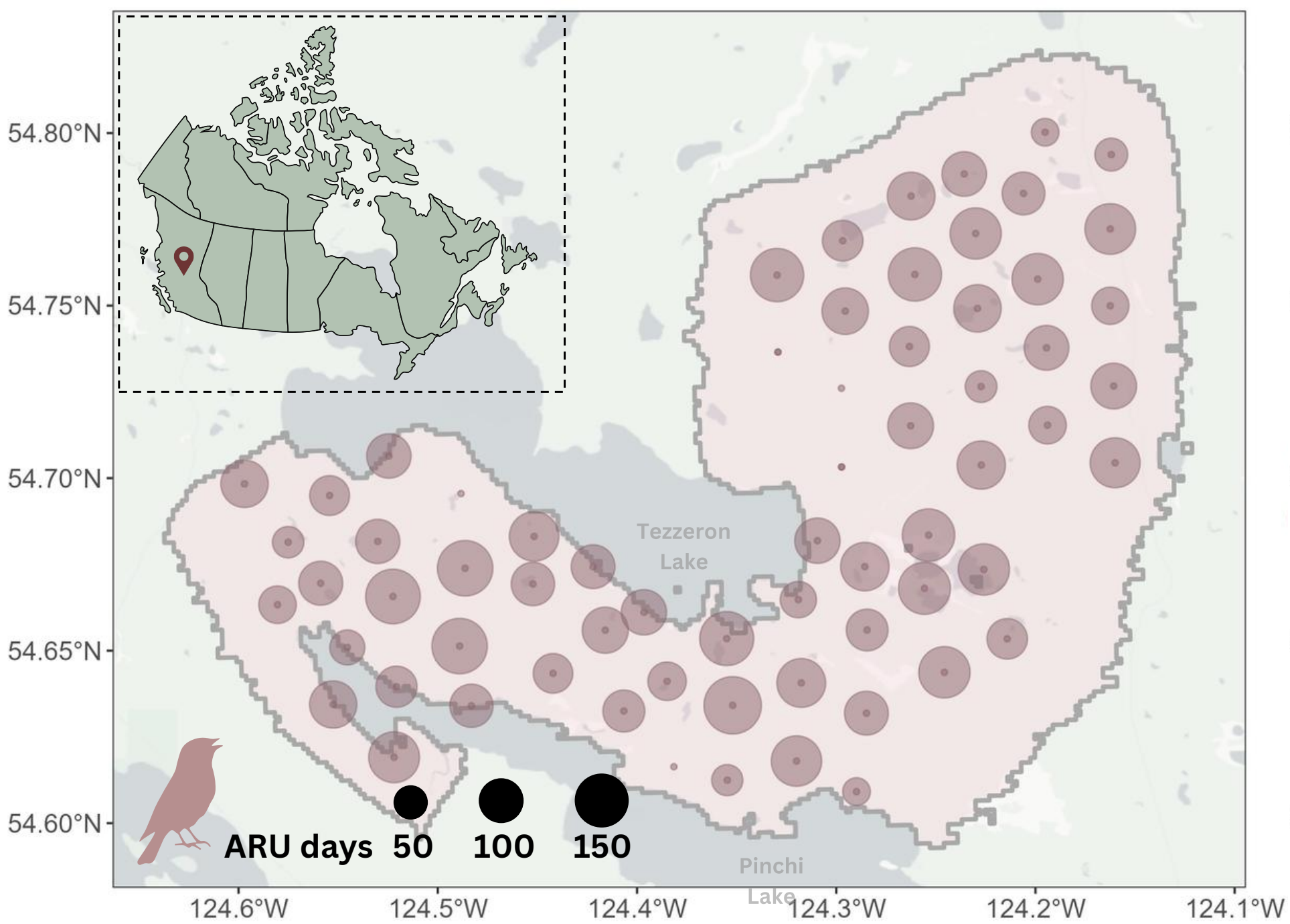


Society of Canadian Ornithologists
Société des ornithologistes du Canada



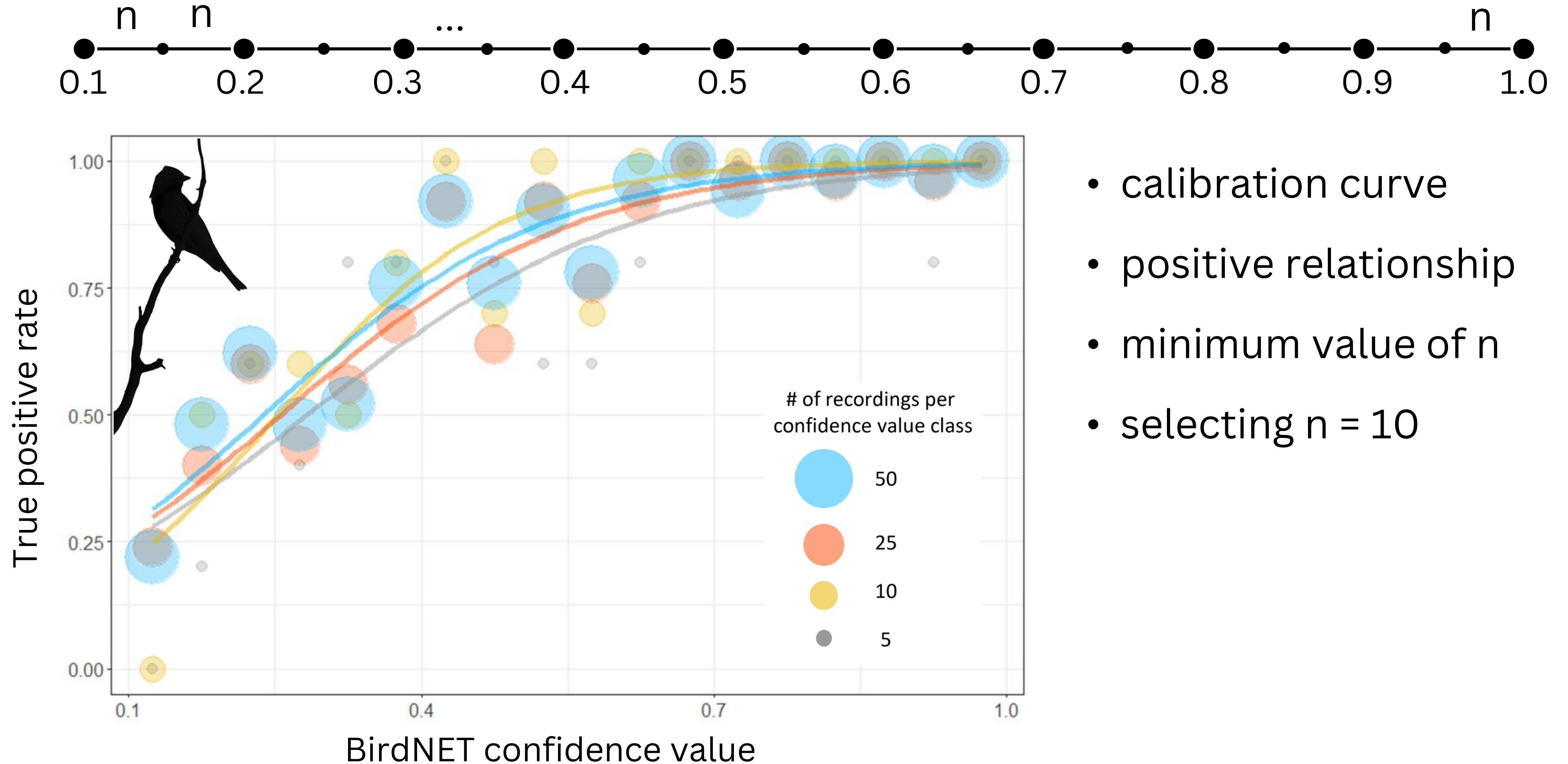
HABITAT CONSERVATION
TRUST FOUNDATION

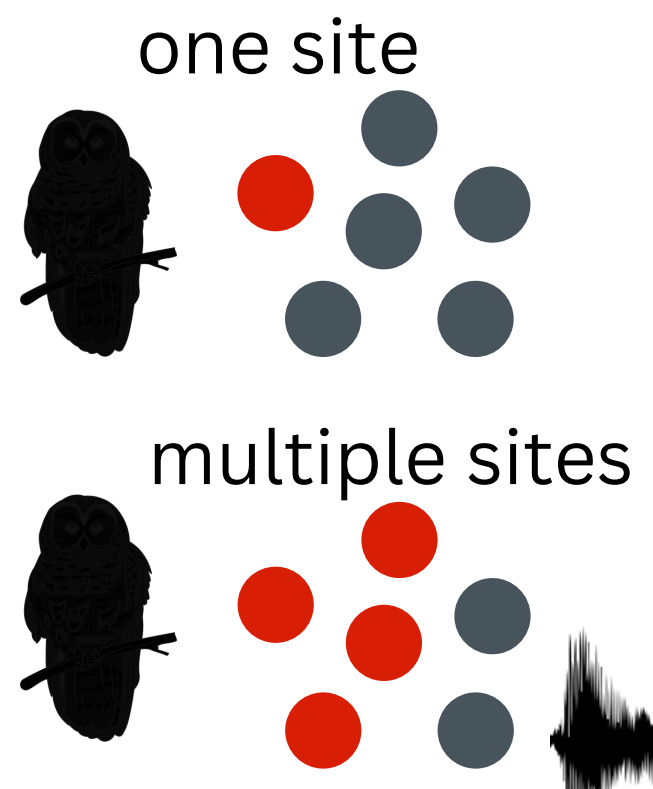
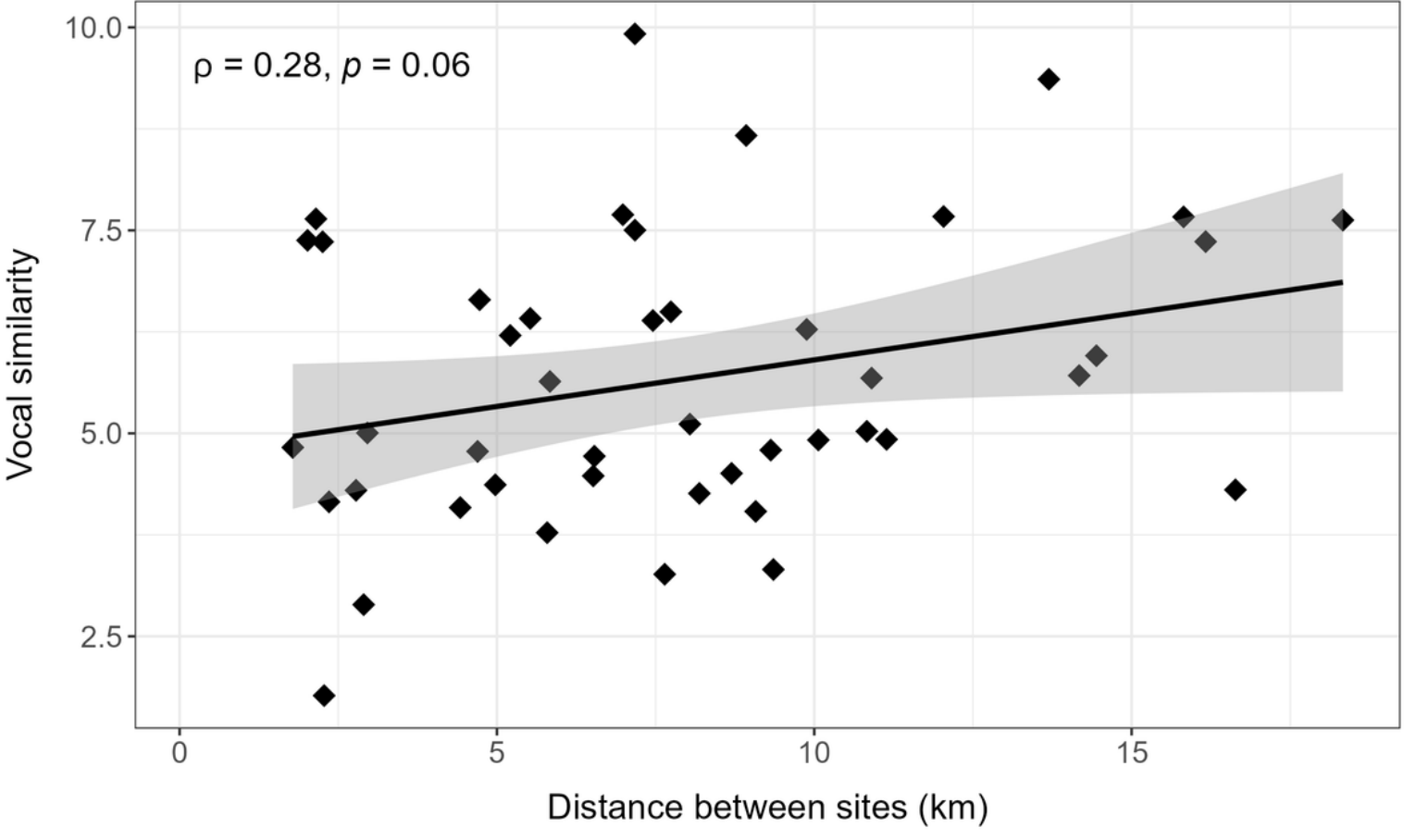
Appendix



Translating BirdNET confidence to probability

Stratified sampling -- balanced evaluation dataset



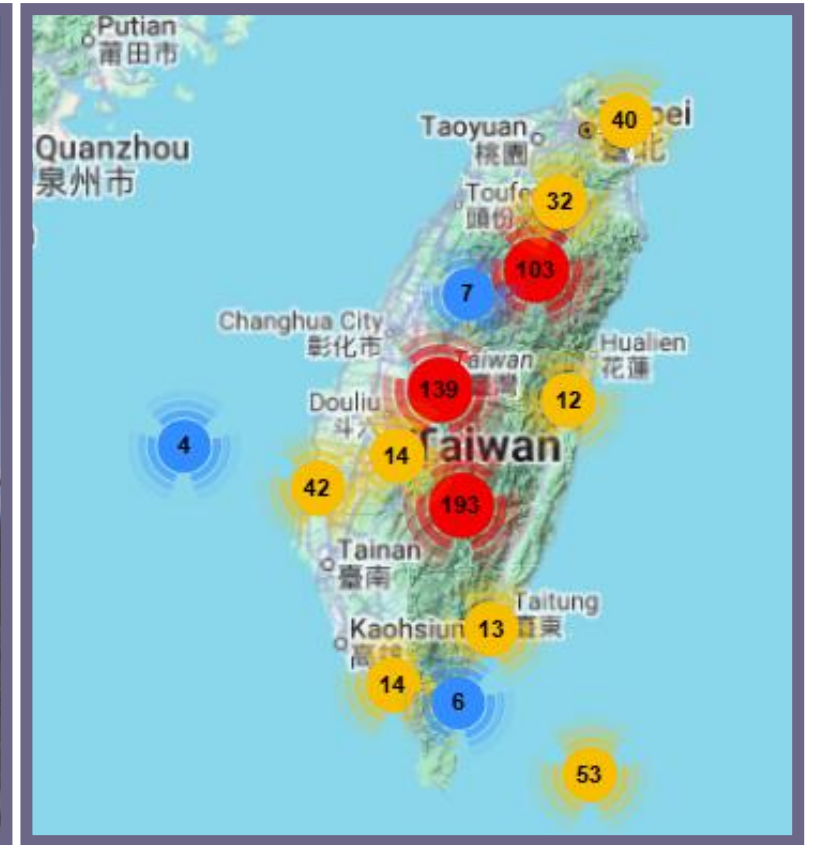




Fieldwork, Siberian Arctic tundra



Endemic Species Research Institute, Taiwan



MSc in UBC, Williams Lake



PhD in UNBC, interior BC

